

A Market-Based Cost of Capital

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Motivation

One common definition of the cost of capital (CoC)

$$\text{CoC} = \text{Expected long-run return}$$

Vital input in financial and economic analysis

- Researchers: discount cash flows and discipline models
- Firms and investors: make investments and allocate capital

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This paper: New method that estimates expected long-run return out of sample and calculates the cost of capital

This Paper: A Market-Based Cost of Capital

1. New method for predicting long-run returns

- Predict cash flows (CF) and infer expected returns (ER) from prices
- Better than direct ER prediction out-of-sample
 - Why? Model for CF more stable than model for ER

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- Global sample of $\sim 16,000$ firms across 93 countries from 1950 to 2023
- Predict firm cash flows using LightGBM
- Outcome: out-of-sample estimates of long-run exp. returns (firm-level)
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- Lower MSE than competing alternatives

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4. A “market-based cost of capital” (MCC)

- Firms whose perceived CoC is closer to the MCC have higher market values

Existing Literature

1. Return expectations

- Expectations deviate from rational benchmarks ([Greenwood and Shleifer 2014](#); [Coibion and Gorodnichenko 2015](#); [Bordalo et al. 2020](#); [Bianchi, Ludvigson, and Ma 2022](#); [Jensen 2024](#); [Bordalo et al. 2024](#); [Bianchi et al. 2025](#))
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2. Long-run stock returns

- Does long-run predictability exist, and does it work out of sample? ([Welch and Goyal 2008](#); [Keloharju, Linnainmaa, and Nyberg 2021](#); [van Binsbergen et al. 2023](#); [Cho and Polk 2024](#); [Liu, Xue, and Zhang 2026](#))
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3. Finance in Practice

- How should firms and investors calculate CoC? How do they? ([Stein 1996](#); [Graham and Harvey 2001](#); [Décaire and Graham 2024](#); [Gormsen and Huber 2025](#); [Hommel, Landier, and Thesmar 2025](#))
- This paper: an unbiased cost of capital firms can use — and evidence that using it is associated with higher market value

Methodology

Two Approaches to Estimating Long-Run Returns

- Two-period equity claim:

$$P_t = \frac{CF_{t+1} + P_{t+1}}{1 + R_{t+1}} = (1 + PD_{t+1}) \frac{CF_{t+1}}{1 + R_{t+1}}$$

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1.b In samples with n non-overlapping periods:

$$\text{MSE}_n^{ER} - \text{MSE}_n^{CF\text{-simple}} \approx \left(1 + \frac{4}{n}\right) \text{var}(\beta_t^p) + \frac{2}{n} \text{var}(pd_{t+1})$$

Predicting Cash Flows

- We estimate a model with the form

$$y_{t+h}^i = f(X_t^i)$$

where

X_t 382 firm characteristics from accounting, price, analyst, industry, and country data based on [Jensen, Kelly, and Pedersen \(2023\)](#)

$f(\cdot)$ A LightGBM machine learning model

y_{t+h} Realized CF at year $t + 1, \dots, t + 10$ (real NI and FCFE)

- Sample: data pooled across 93 countries (U.S. 1950–2023; non-U.S. 1985–2023)
- Out-of-sample period:
 - US: 1984–2023
 - Non-US: 1996–2023,with parameters re-estimated yearly

From Expected Cash Flows to Expected Returns

10-year explicit horizon, terminal value grows at constant rate g_t

$$p_t = \sum_{h=1}^{10} \frac{\mathbb{E}_t[\text{CF}_{t+h}]}{(1 + r_t^{\text{equity}})^h} + \frac{\mathbb{E}_t[\text{CF}_{t+10}](1 + g_t)}{r_t^{\text{equity}} - g_t} \frac{1}{(1 + r_t^{\text{equity}})^{10}}$$

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- g_t : country-specific forward real GDP growth (IMF)
- Solve for $r_t^{\text{equity}} \in (g_t, 10\%)$
- Final estimate: average of the NI- and FCFE-based estimates

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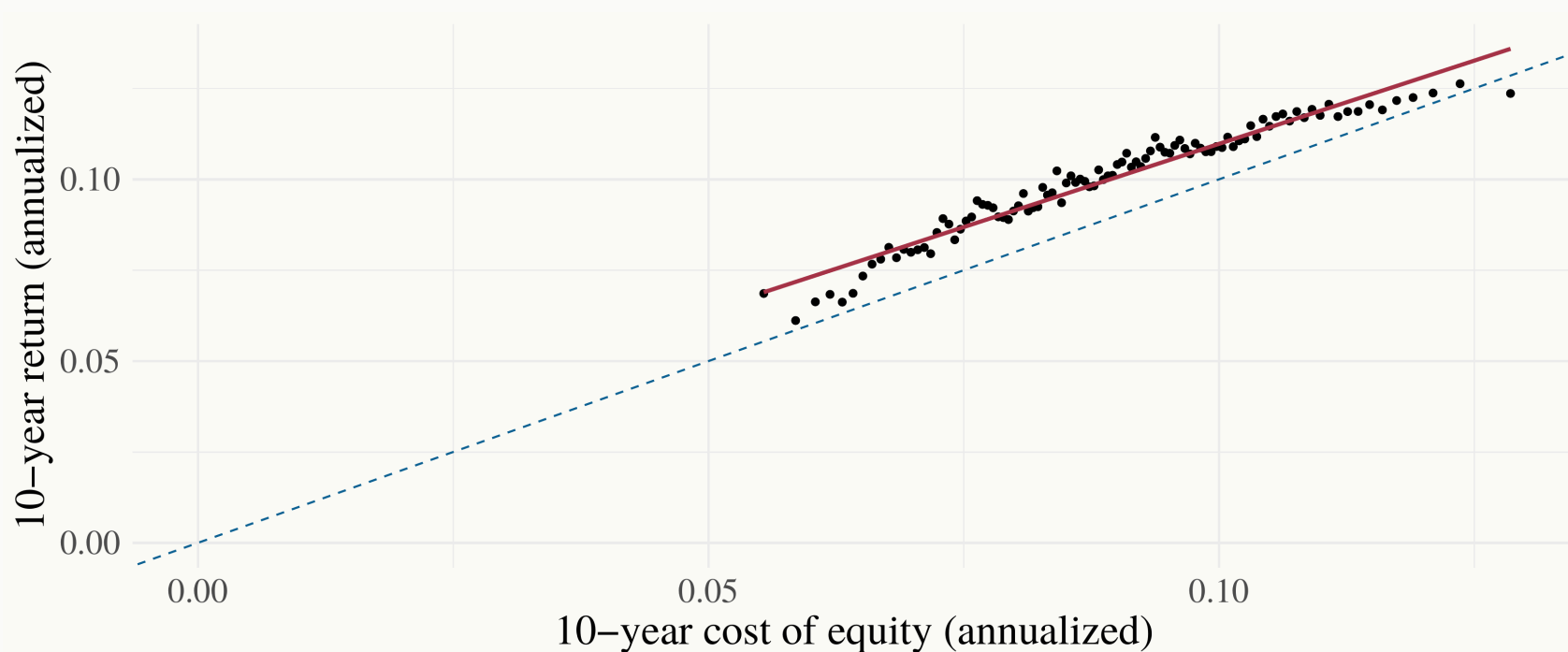
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⇒ **The market-based cost of equity (MCE)**: the expected return that equates discounted expected cash flows with the observed market value

Testing the MCE

The MCE: an Unbiased Estimate of Returns

Each month, stocks are sorted into 100 portfolios on the MCE; realized returns over the next 10 years



Global stocks, 1983–2023. Both axes annualized.

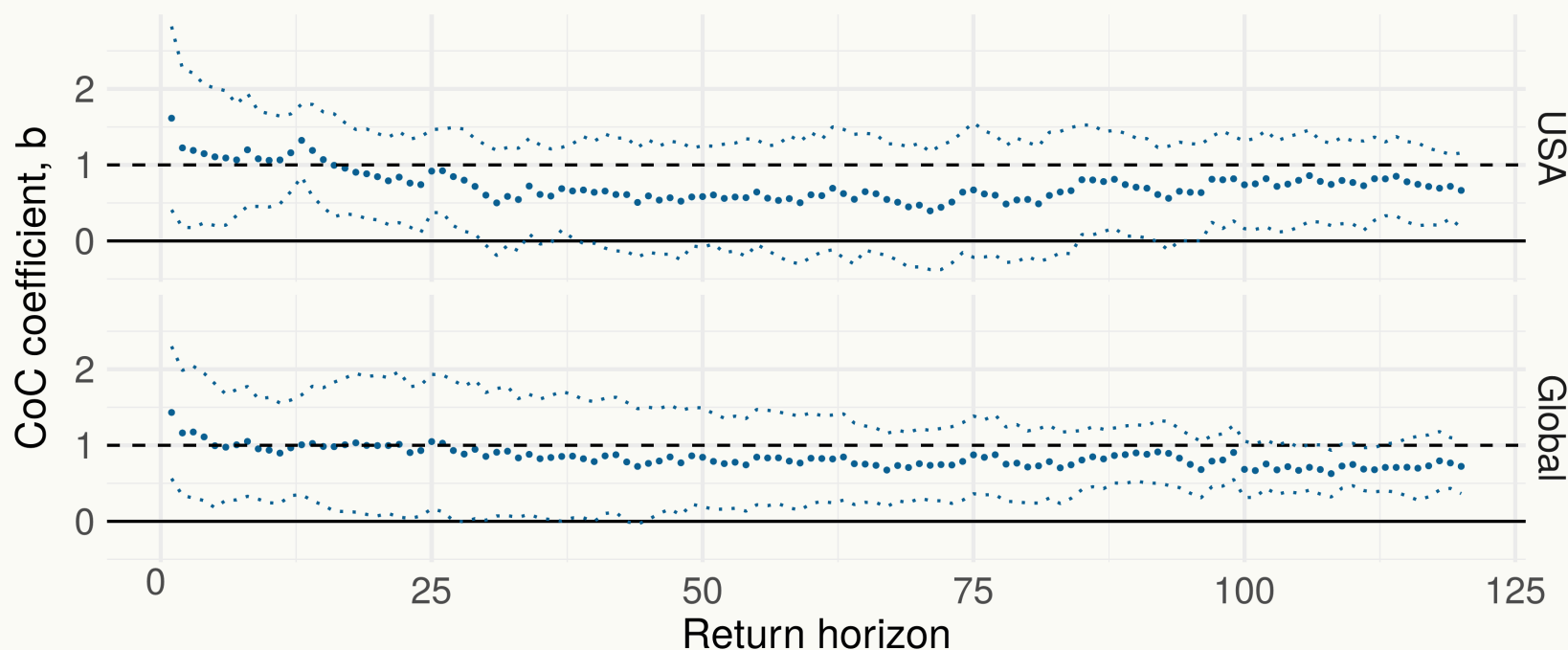
The MCE: an Unbiased Estimate of Returns

Regressions of 10-year realized returns on the 10-year MCE (month fixed effects)

	10-year realized return		
	U.S. (i)	Global (ii)	Global ex-U.S. (iii)
<i>Panel A: Regression estimates</i>			
Cost of equity	0.93*** (0.08)	0.96*** (0.18)	1.01*** (0.23)
Observations	1,106,464	2,610,489	1,504,025
<i>Panel B: Predictive accuracy ($\alpha = 0, \beta = 1$)</i>			
Avg. error	0.25 (0.12)	0.14 (0.10)	0.06 (0.15)
OOS R^2_{CAPM} (%)	2.1	3.8	5.3
OOS R^2_{MKT} (%)	0.4	2.1	4.1
OOS R^2_{OWN} (%)	9.0	12.1	14.7

Forward Return Predictability

Regressions of the return in month $t + h$ on the monthly MCE, $h = 1, \dots, 120$ (month fixed effects)



Dotted lines: 95% confidence intervals, Driscoll-Kraay standard errors

The MCE Also Predicts Market and Industry Returns

Value-weighted portfolios rebalanced monthly

Regressions of 10-year realized portfolio returns on the portfolio MCE

	10-year realized return			
	Country		Industry	
	(i)	(ii)	(iii)	(iv)
Cost of equity	1.40** (0.56)	1.21** (0.47)	1.30*** (0.10)	0.74*** (0.22)
Month FE	No	Yes	No	Yes
Observations	6,433	6,433	14,136	14,136
R ²	0.089	0.237	0.086	0.171

Country portfolios (value-weighted) and Fama-French 49 industry portfolios formed monthly from the global sample. Portfolio MCE solved from the aggregated cash flows and prices of member firms. Country-month observations weighted by the country's share of firms. Formation months 1984–2013. Driscoll-Kraay standard errors.

The MCE vs Alternatives

Machine-learned returns (MLR): predict returns directly

- Same LightGBM model and predictors as the MCE — only the target differs: 10-year returns instead of cash flows
- The empirical counterpart of the ER approach

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Implied cost of capital (ICC): back out expected returns from analyst forecasts

- CT (Claus and Thomas 2001), GLS (Gebhardt, Lee, and Swaminathan 2001), PEG (Easton 2004), OJ (Ohlson and Juettner-Nauroth 2005)
- Same valuation logic as the MCE — but cash flow forecasts from analysts

The MCE Versus Alternatives

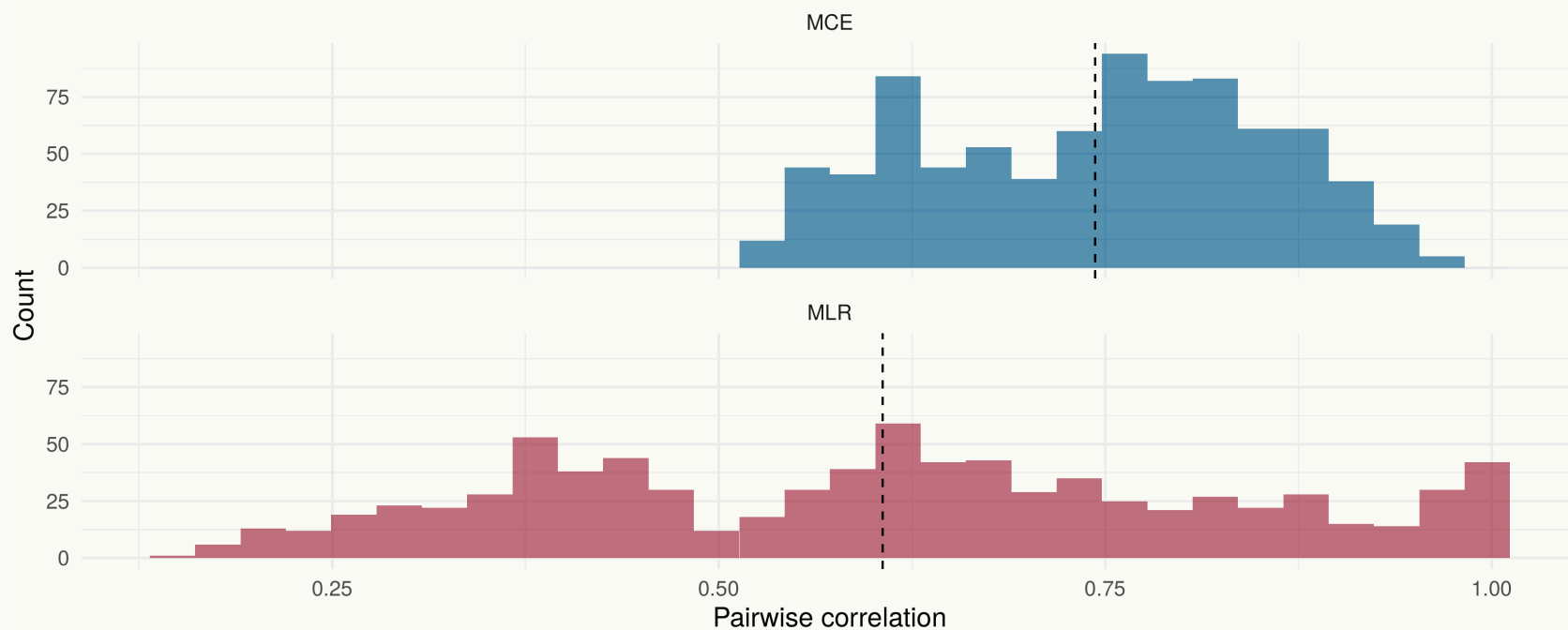
Regressions of 10-year realized returns on each cost of equity estimate (month FEs)

	LightGBM		Implied Cost of Capital			
	MCE	MLR	CT	GLS	PEG	OJ
Slope	0.96*** (0.18)	0.50*** (0.16)	0.11 (0.13)	0.38** (0.17)	0.10 (0.10)	0.07 (0.10)
Δ MSE relative to MCE	—	0.94	0.25	0.12	0.34	0.88

- Only the MCE has a slope near 1, and all alternatives have larger MSE

The MCE Beats MLR: Model Stability

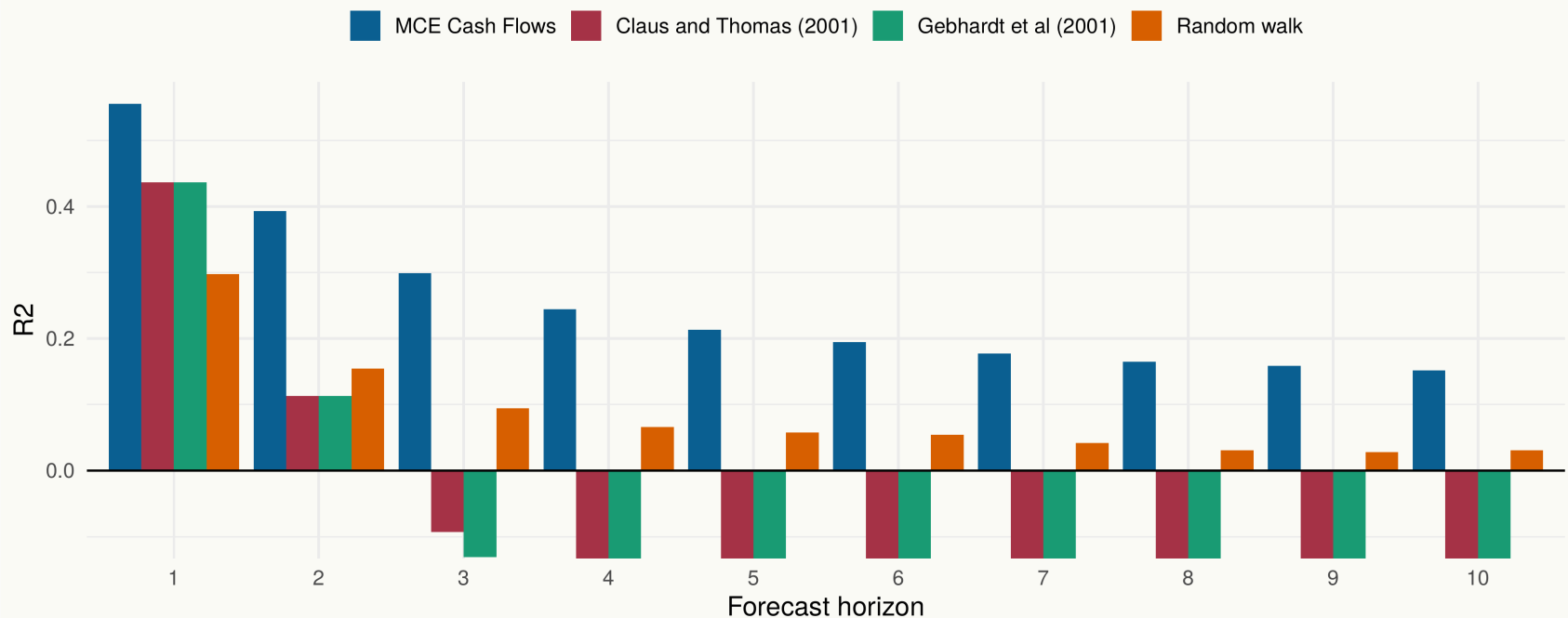
Predictions from every training vintage (1983, ..., 2023) applied to the same stocks;
pairwise correlations between vintages



MCE predictions are similar across model vintages; MLR predictions are not
⇒ **the cash flow model is more stable**

The MCE Beats ICC: Better Cash Flow Estimates

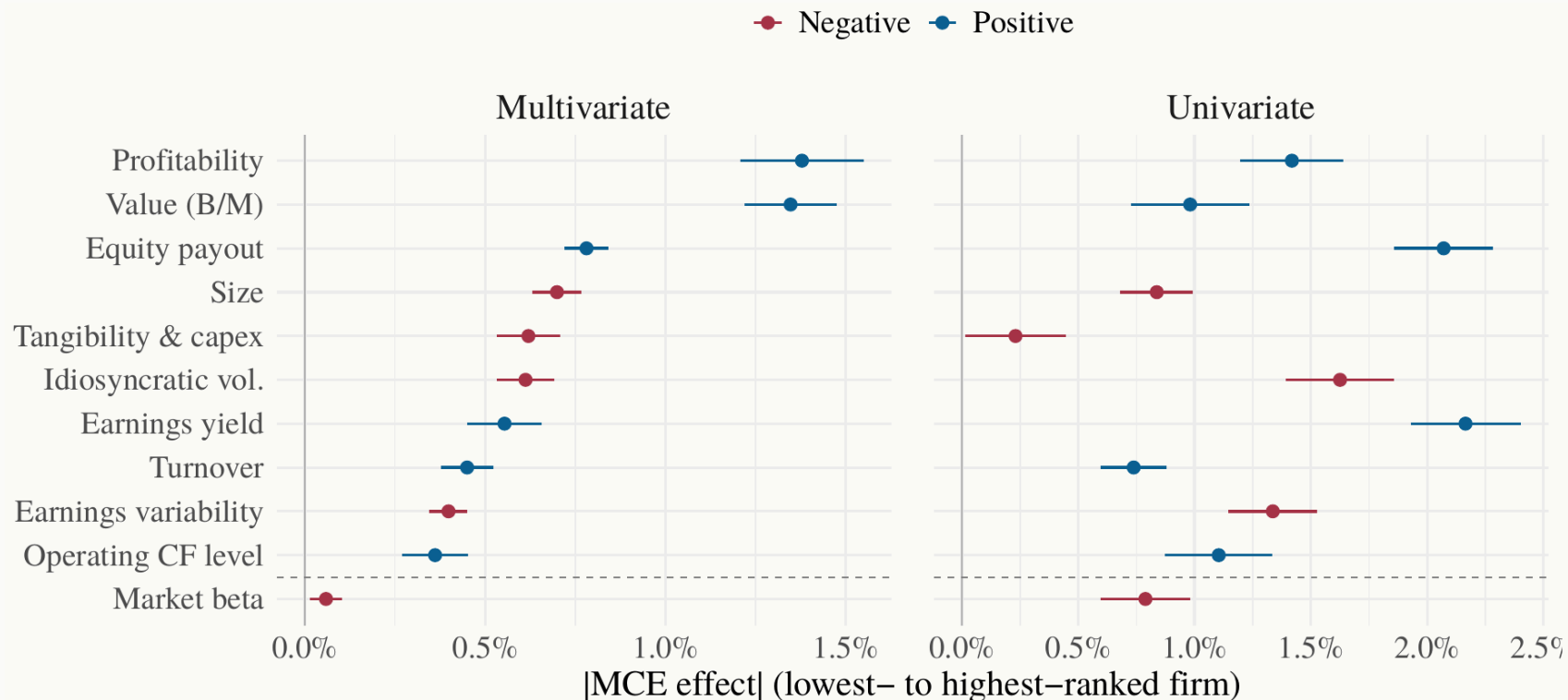
Out-of-sample R^2 for net income forecasts at horizons 1–10 years



Inside the MCE

Which Characteristics Matter?

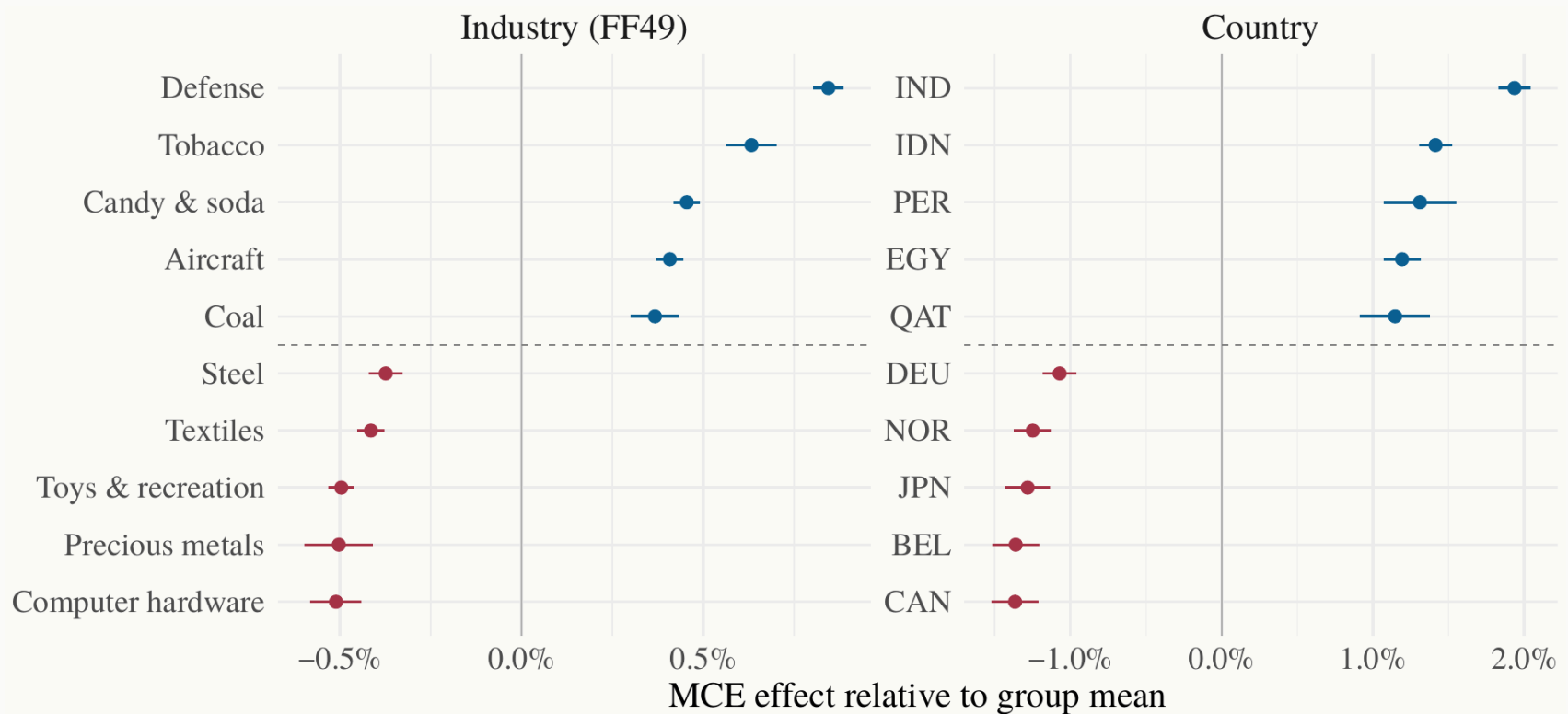
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Absolute coefficients, colour is the sign; 95% CIs, SEs clustered by industry and month

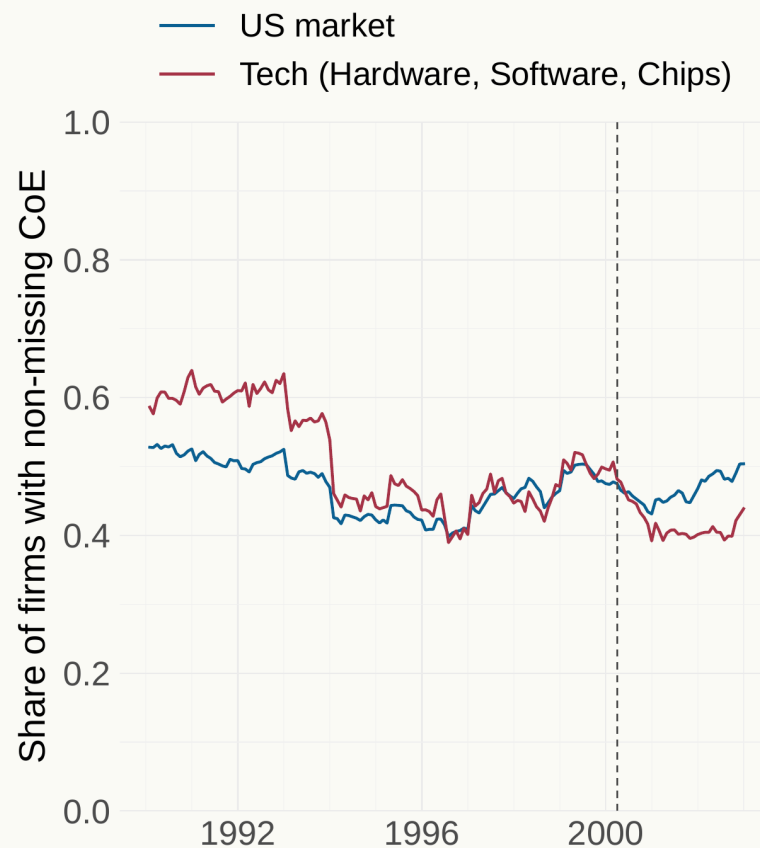
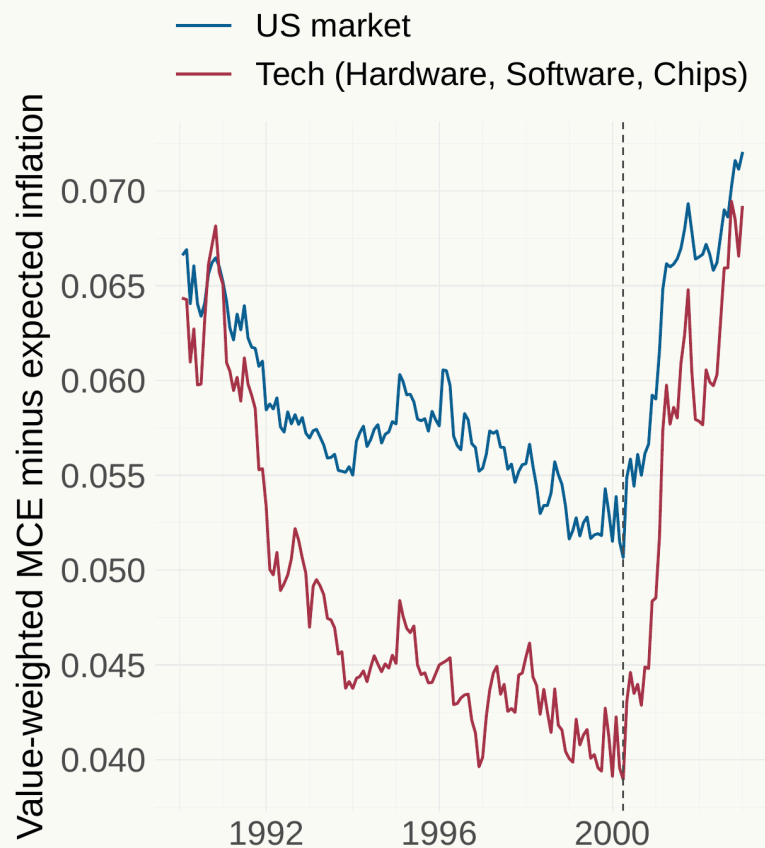
- High MCE: profitable, value, high-payout, and small firms
- Low MCE: capex-heavy and volatile firms

Which Industries and Countries? Top and Bottom 5



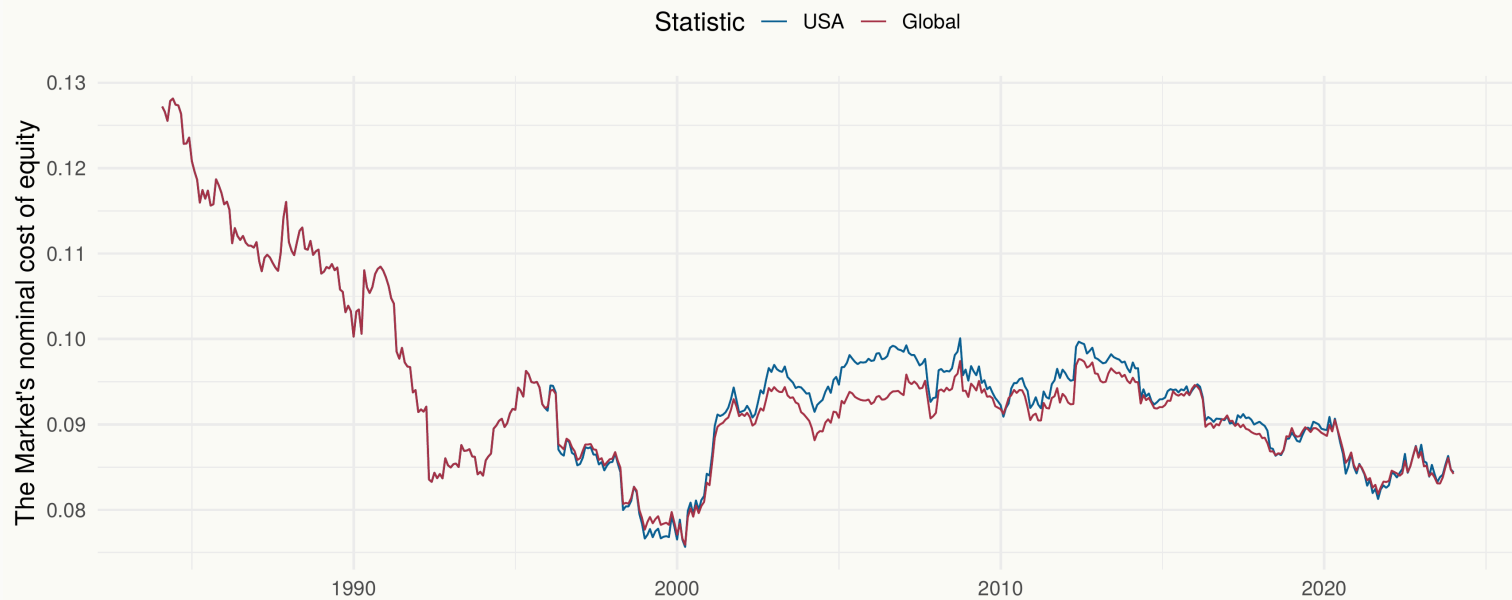
Dummies from the multivariate regression, as deviations from the group mean; 95% CIs

The Dot-Com Bubble: Tech Firms vs. the U.S. Market



Tech: FF49 Hardware, Software, and Chips. Left: value-weighted MCE minus expected inflation. Right: share of firms with an MCE. Dashed line: March 2000.

Time Variation in the Market's Nominal MCE



Does the MCE Capture Risk or Expectation Errors?

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Panel regressions of risk and mispricing proxies on the MCE

- Risk: analyst risk ratings ([Jensen 2024](#)) and market beta
- Mispricing: earnings announcement returns (EAR) and realized and expected analyst forecast errors (FE, $\mathbb{E}[\text{FE}]$)

	Risk			Mispricing		
	Morningstar	Value Line	Beta	EAR	FE	$\mathbb{E}[\text{FE}]$
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
MCE	-0.08*** (0.02)	-0.22*** (0.01)	-0.13*** (0.01)	0.10*** (0.01)	0.66*** (0.05)	1.08*** (0.05)
Observations	112,979	382,036	375,295	191,573	24,806	24,806
R ²	0.015	0.117	0.038	0.004	0.062	0.177

Time fixed effects; risk proxies standardized; SEs clustered by firm and time

The Market-Based Cost of Capital

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Insert the MCE into the weighted average cost of capital

$$\text{MCC} = \frac{D}{D+E} (1 - \tau) r_t^{\text{debt}} + \frac{E}{D+E} r_t^{\text{equity}}$$

- r_t^{equity} : the MCE
- r_t^{debt} : bond yields net of expected default losses
 - Projected on distance-to-default for firms without traded bonds
- Leverage from market equity and book debt; τ from effective tax rates

Using a Market-Based Cost of Capital

Standard theory: firms maximize market value by using expected returns as CoC

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Gormsen & Huber (2025)

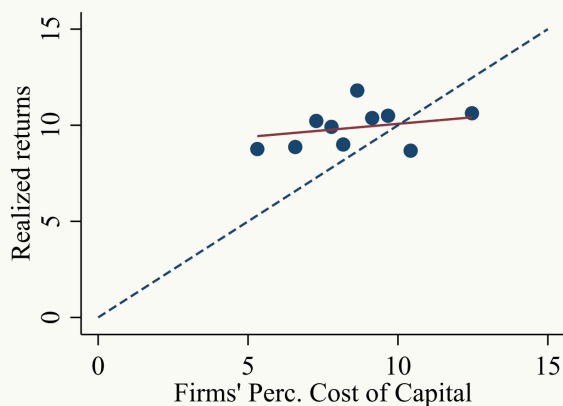
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- Firms' perceived CoC deviates from expected returns
- One interpretation: firms do not aim to maximize market value
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Realized returns on 10 portfolios sorted on perceived CoC, global sample, 2002–2024

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Using a Market-Based Cost of Capital

Standard theory: firms maximize market value by using expected returns as CoC

Gormsen & Huber (2025)

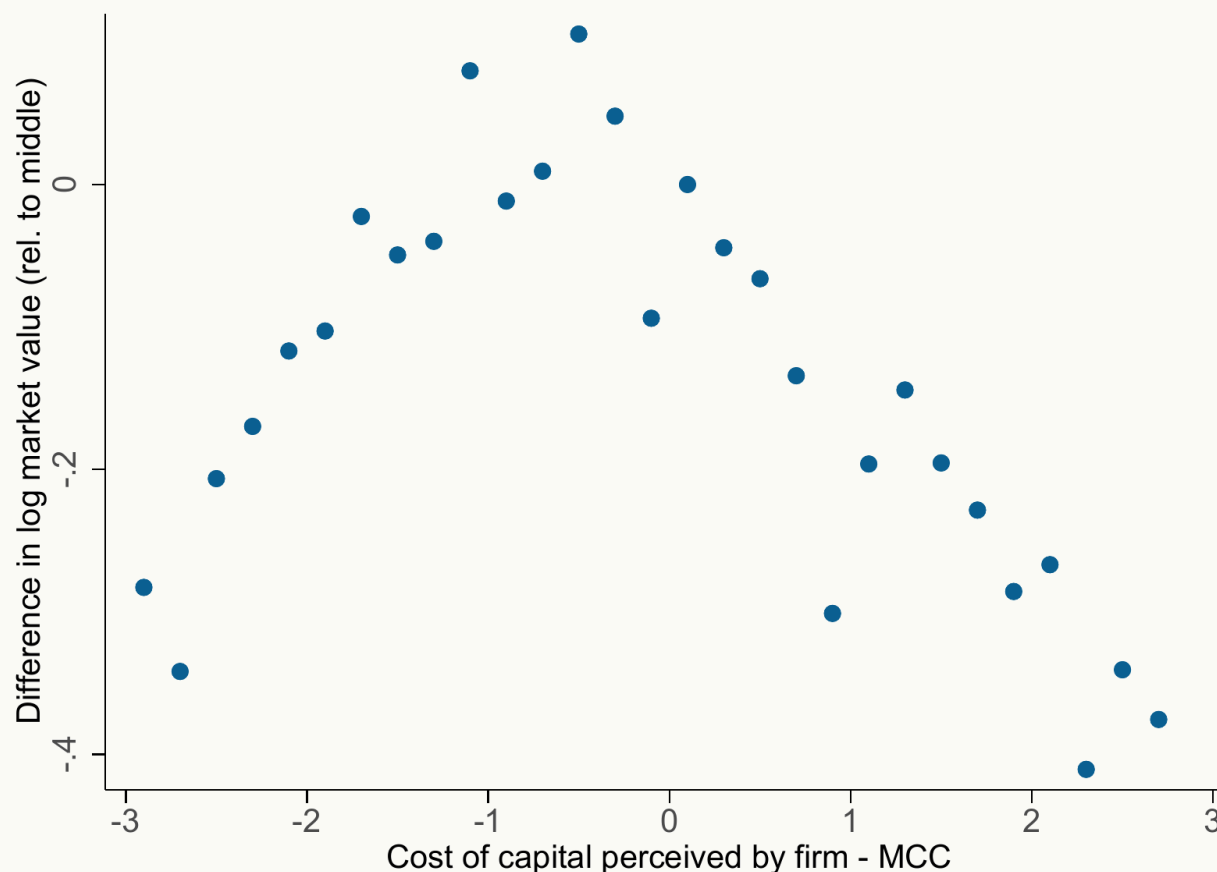
- Studies firms' perceived cost of capital
- Firms' perceived CoC deviates from expected returns
- One interpretation: firms do not aim to maximize market value
- Alternative interpretation: some firms aim to maximize market value but don't know expected returns

How should firms perc. CoC relate to market value?

- Firms with a perceived CoC closer to the MCC have higher market values
- Deviations destroy value: too low $\Rightarrow$ overinvest, too high $\Rightarrow$ underinvest

The Inverted U: Market Value and Deviations from the MCC

Log market value on bins of the cost of capital deviation (perc. CoC — MCC)



Book equity decile and quarter-by-country fixed effects

The Inverted U: Regression Evidence

Log market value on the absolute CoC deviation (1–2) and on the deviation by region (3–5)

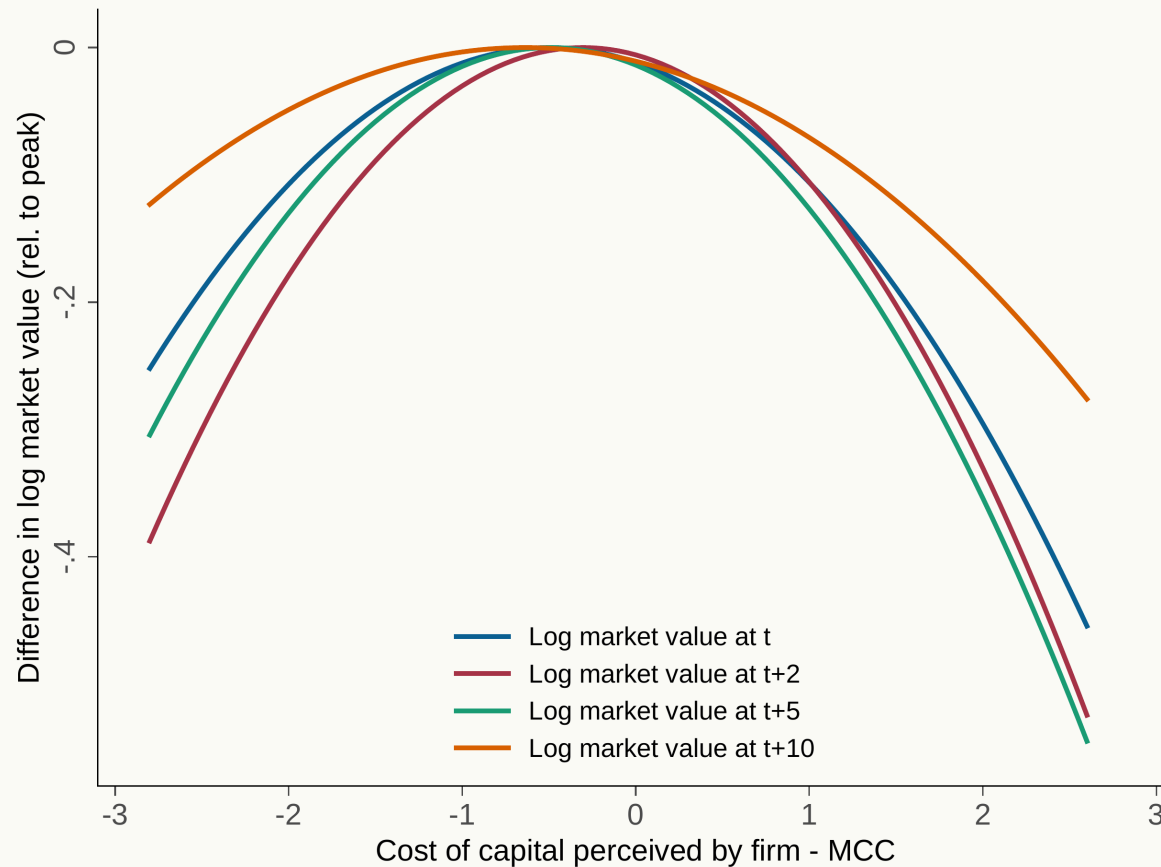
	(1)	(2)	(3)	(4)	(5)
	U.S.	Global	≤ -0.3	≥ 0.3	$> -0.3, < 0.3$
Perc. CoC — MCC	-0.081*** (0.019)	-0.096*** (0.018)			
Perc. CoC — MCC			0.087* (0.049)	-0.070*** (0.025)	-0.173 (0.210)
Observations	3,098	5,204	1,509	3,055	640
Within R ²	0.0185	0.0266	0.00924	0.0154	0.00240

Book equity decile and quarter-by-country fixed effects

Deviations and Future Changes in the MCC

	(1)	(2)
	$\Delta \text{MCC}_{t \rightarrow t+2}$	$\Delta \text{MCC}_{t \rightarrow t+5}$
Perc. CoC — MCC	0.119*** (0.029)	0.135*** (0.045)
Observations	4,125	3,090
Within R ²	0.042	0.038
Quarter-year $\times$ Country FE	Yes	Yes
Clustering	Firm, date	Firm, date

Deviations and Future Changes in the MCC



Stock Returns and Investment Announcements

Abnormal returns around new investment announcements on the CoC deviation

	(1)	(2)	(3)	(4)
	U.S.	Global	U.S.	Global
(Perc. CoC — MCC) > 0	0.199** (0.100)	0.199** (0.098)		
Perc. CoC — MCC			0.032* (0.019)	0.034* (0.019)
Observations	414	709	414	709
Within R ²	0.0121	0.0107	0.00472	0.00465

Book equity decile and quarter-by-country fixed effects

Conclusions

New method to calculate expected returns and firms' cost of capital: **the MCC**

- Theoretically preferable in setting with stable cash flows and volatile asset pricing dynamics
- Works in practice: unbiased OOS estimates and lower MSE than alternatives

The MCC can be used by

- **firms** that wish to maximize current market value
- **investors** that make long-run capital allocation decisions or valuations
- **researchers** that need unbiased estimates of expected long-run returns

Bonus: Method produces unbiased estimates of expected cash flows

Data to be shared soon!

Thank You!

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Appendix Contents

A.1 Return Predictability

A.2 MCE vs. Alternatives

A.3 Cash Flow Predictability

A.4 Delisting, Imputation, and MCE Validity

A.5 Cost of Capital Estimates

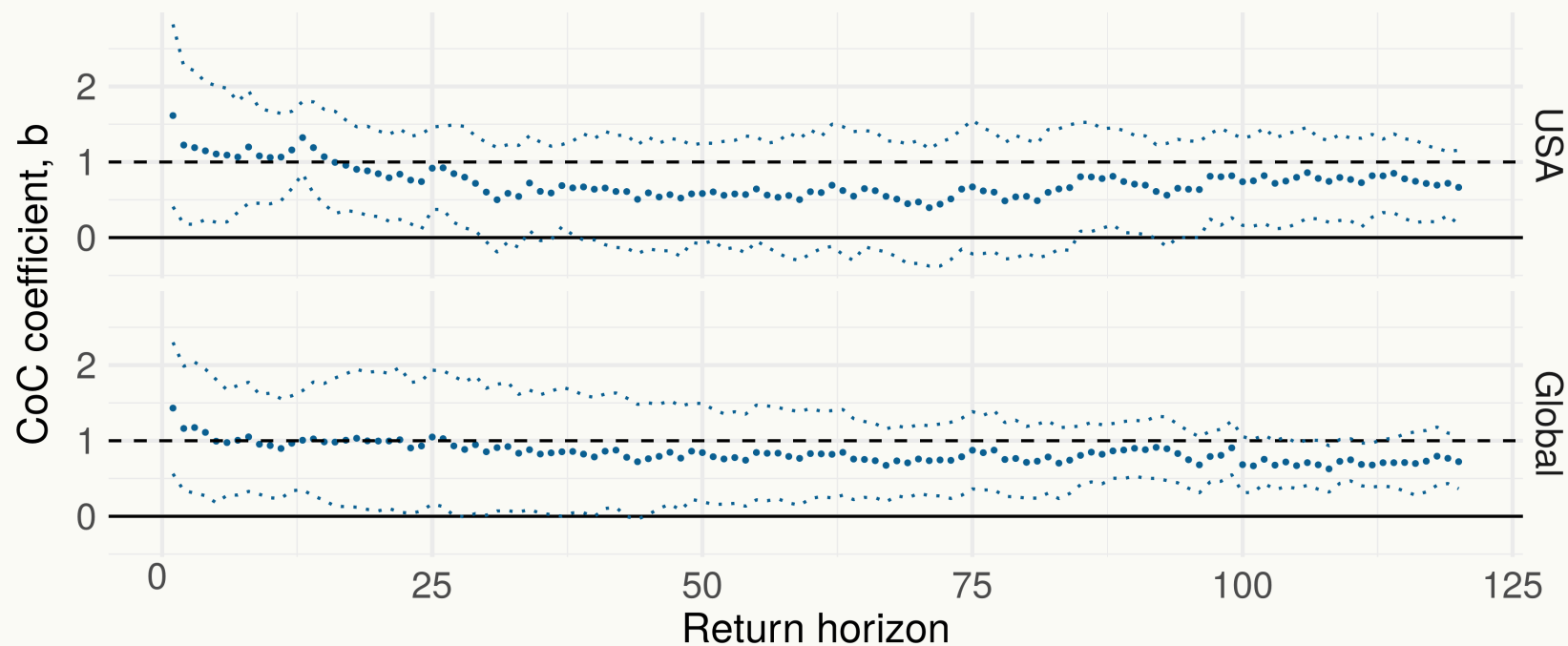
A.6 Risk or Mispricing

A.7 Firm Value and Investment

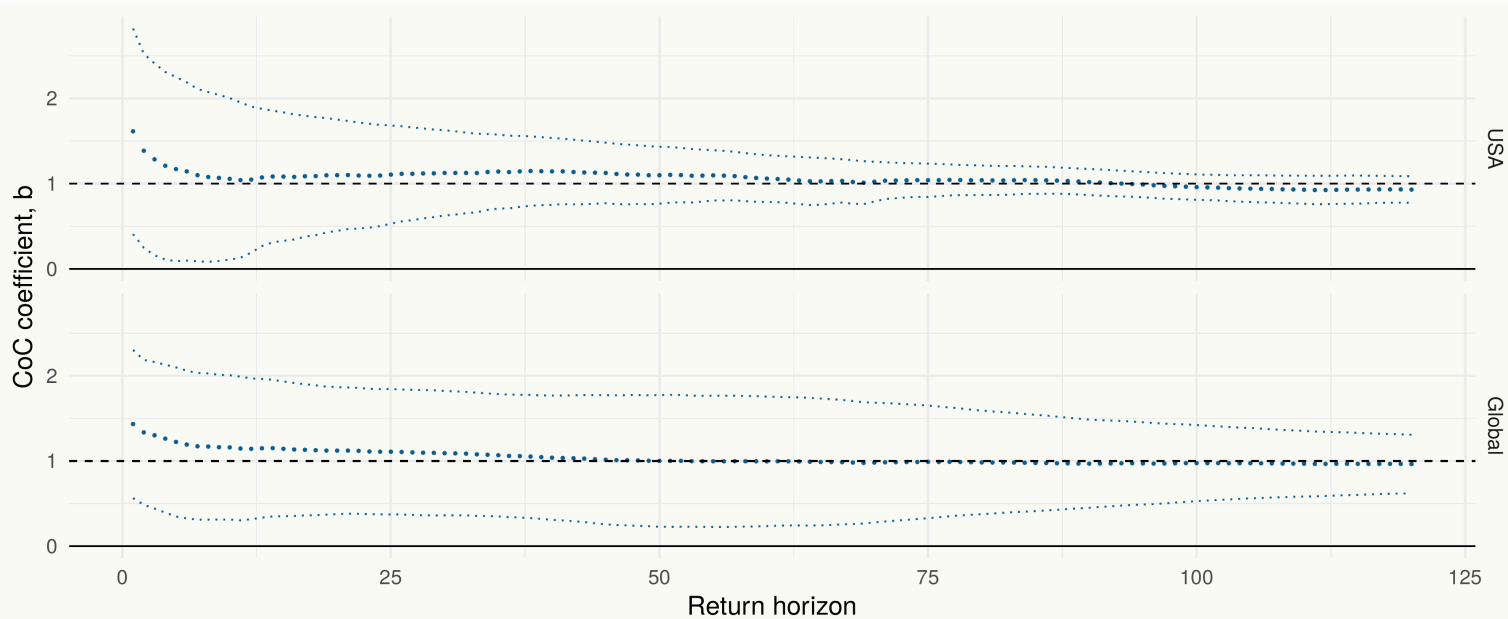
A.8 Model Details

A.1 Return Predictability

Forward Return Predictability



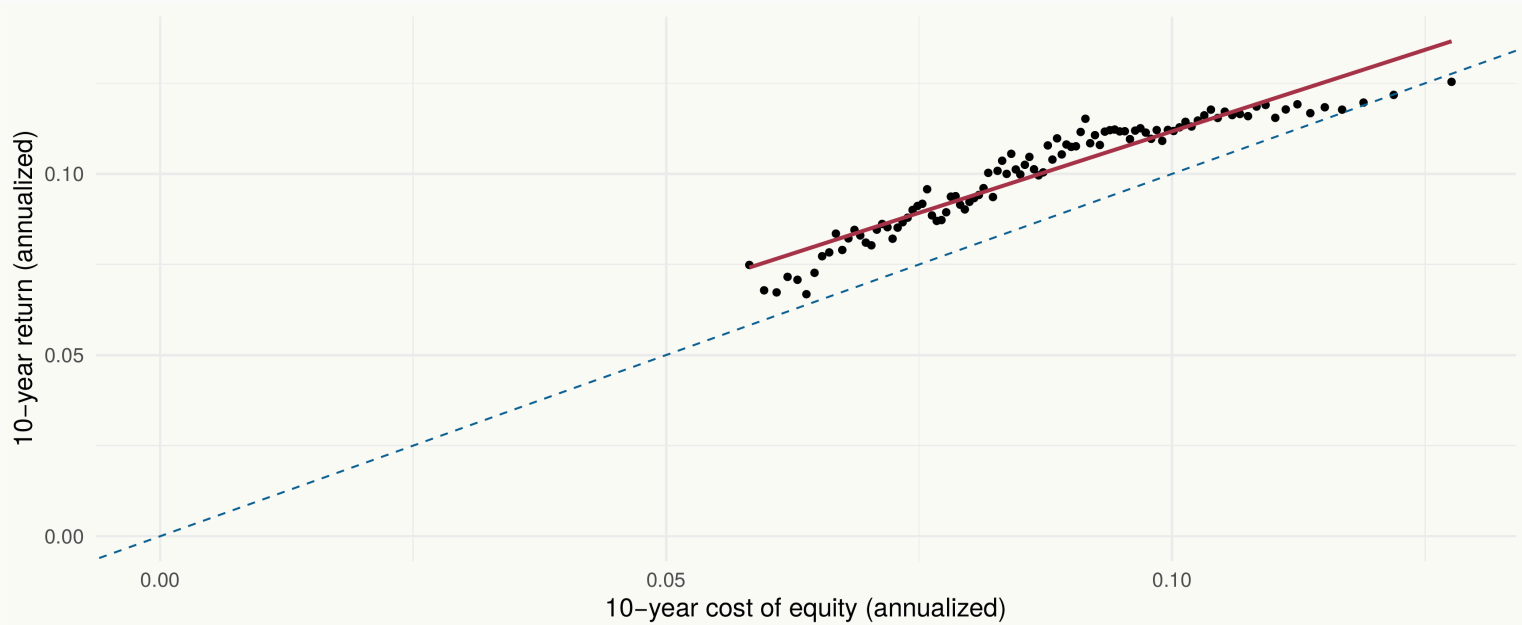
Buy-and-Hold Return Predictability



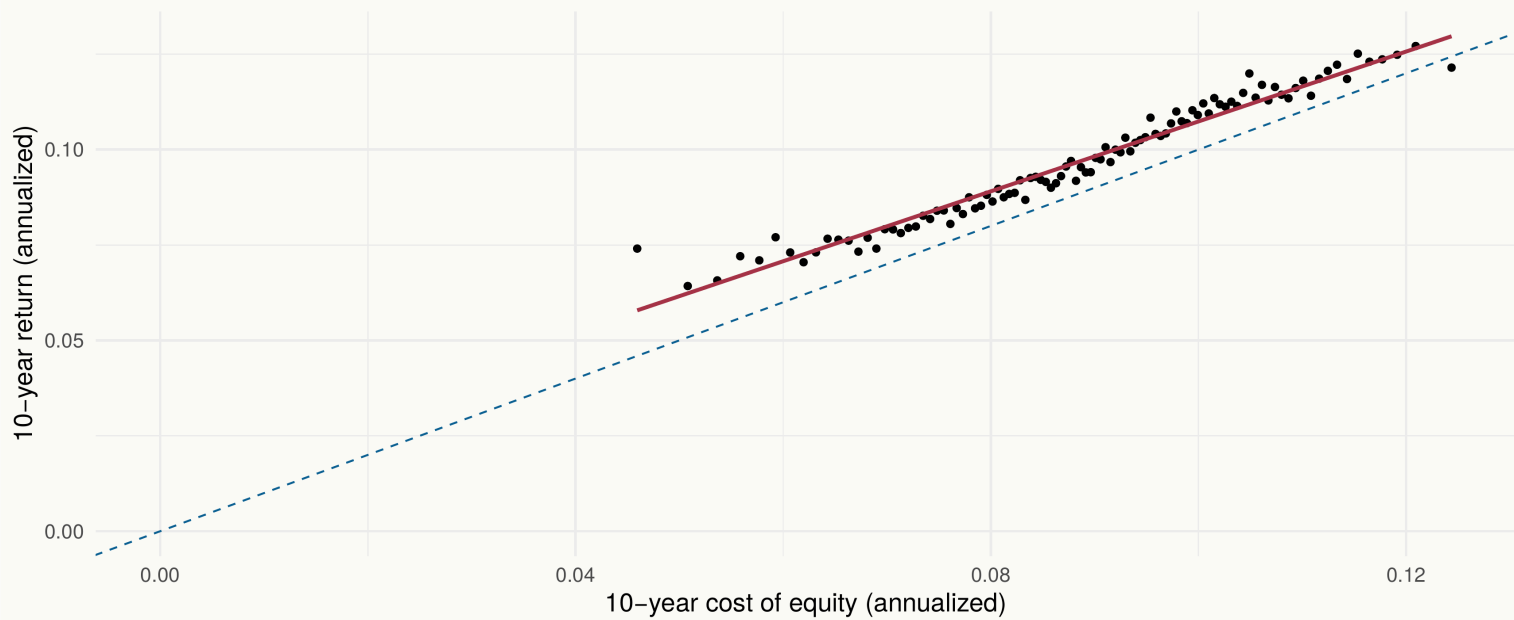
Return Predictability: Fama–MacBeth

	10-year realized return		
	U.S. (i)	Global (ii)	Global ex-U.S. (iii)
Cost of equity	0.93*** (0.08)	0.91*** (0.15)	0.94*** (0.23)
Months	363	363	220

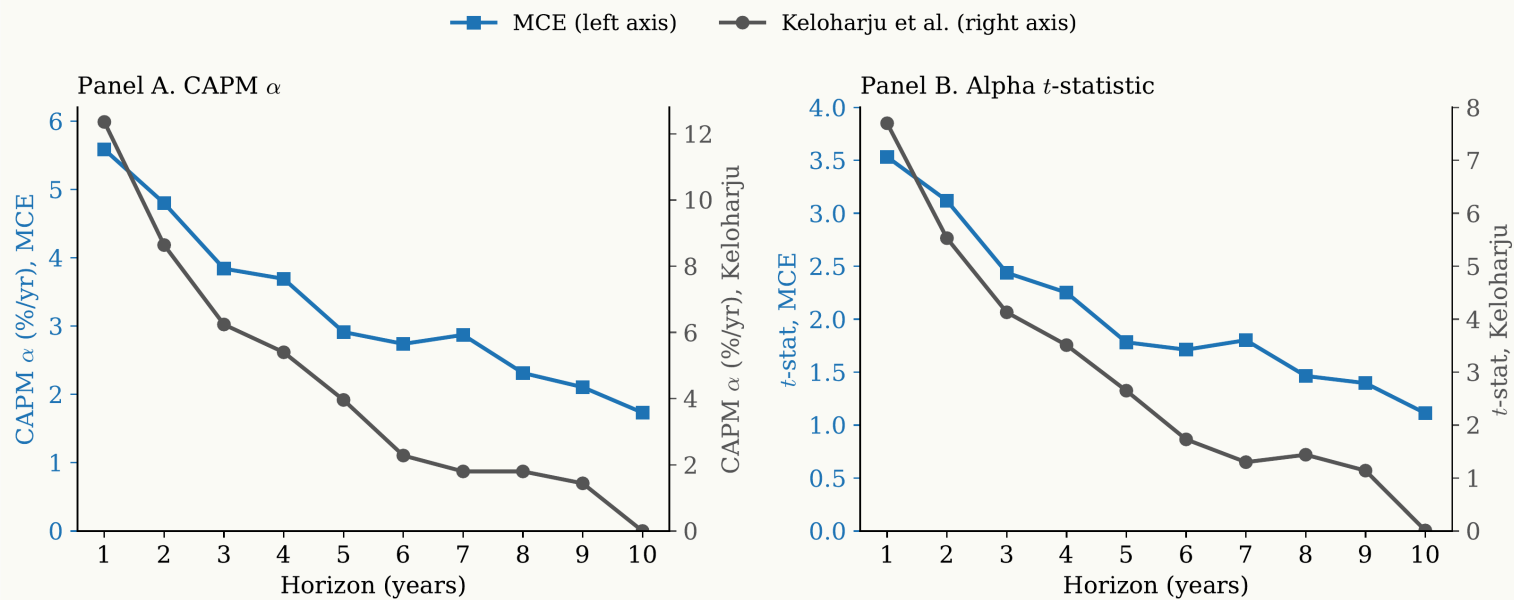
Long-Run Returns and the MCE (U.S. Sample)



Long-Run Returns and the MCE (Non-U.S. Sample)



Forward Return Predictability: MCE vs. Keloharju et al.



A.2 MCE vs. Alternatives

Implied Cost of Capital Measures

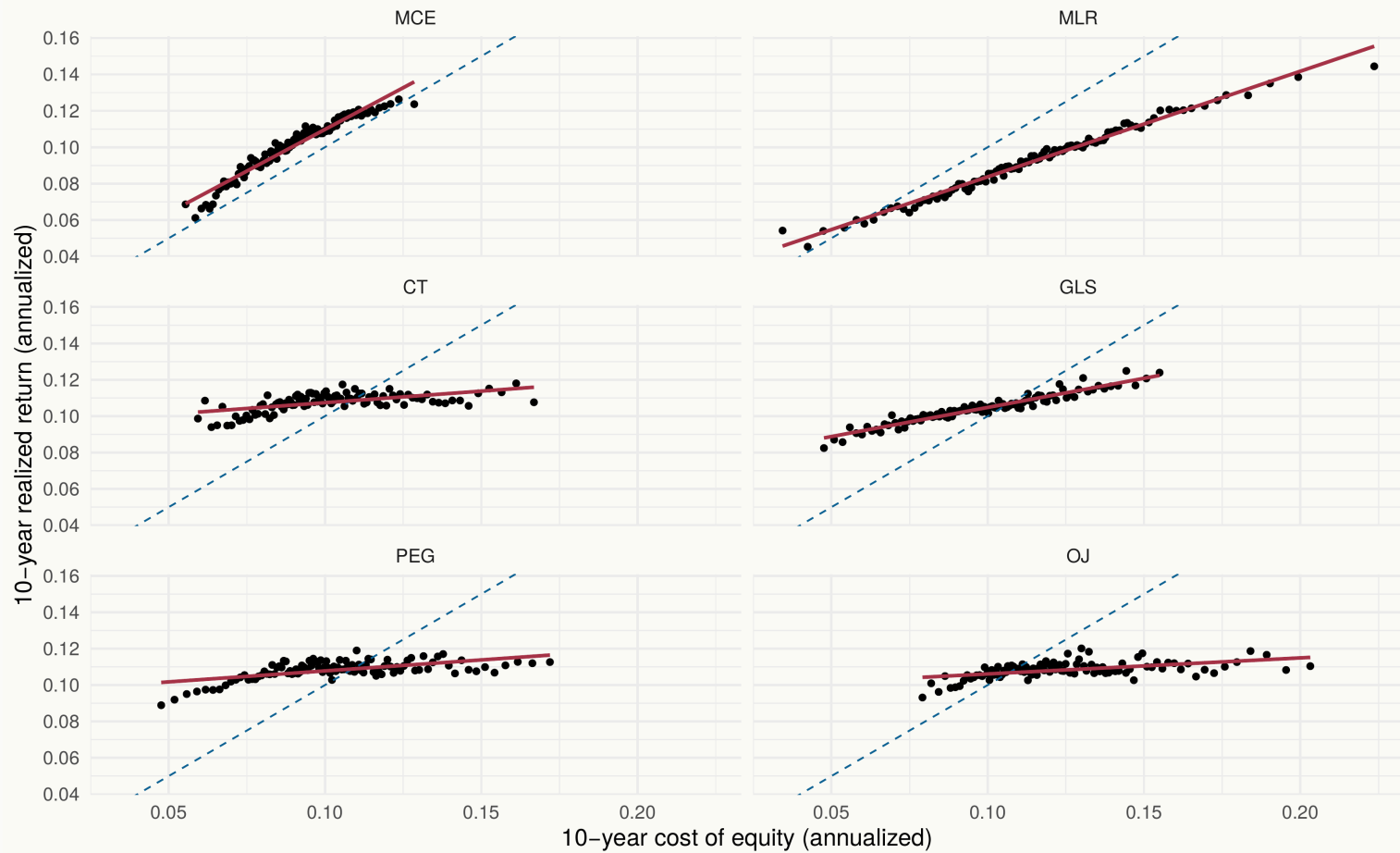
Implied cost of capital (ICC): back out expected returns from analyst forecasts

- Same valuation logic as the MCE — but cash flow forecasts from analysts
- Four standard implementations
 - CT ([Claus and Thomas 2001](#)): 2 years of forecasts, grown with the long-term growth forecast to year 5
 - GLS ([Gebhardt, Lee, and Swaminathan 2001](#)): 2 years of forecasts, then ROE fades linearly to the industry median by year 12
 - PEG ([Easton 2004](#)): 2 years of forecasts + long-term growth forecast
 - OJ ([Ohlson and Juettner-Nauroth 2005](#)): 2 years of forecasts + long-term growth forecast

Return Benchmarks

	10-year realized return					
	LightGBM methods		Implied Cost of Capital methods			
	MCE	MLR	CT	GLS	PEG	OJ
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
<i>Panel A: Regression estimates</i>						
Cost of equity	0.96***	0.50***	0.11	0.38**	0.10	0.07
	(0.18)	(0.16)	(0.13)	(0.17)	(0.10)	(0.10)
Observations	2,610,489	4,872,824	1,003,308	1,877,706	1,000,734	985,775
<i>Panel B: Predictive accuracy ($\alpha = 0, \beta = 1$)</i>						
Avg. error	0.14	-0.19	0.05	-0.00	-0.05	-0.56
	(0.10)	(0.28)	(0.08)	(0.08)	(0.10)	(0.08)
OOS R^2_{MKT} (%)	2.1	-0.5	-0.1	1.1	-0.5	-2.6
OOS R^2_{CAPM} (%)	3.8	0.9	1.4	2.9	1.3	-1.3
OOS R^2_{OWN} (%)	12.1	6.6	14.0	14.1	16.3	12.9
<i>Panel C: MSE relative to MCE</i>						
ΔMSE	—	0.94	0.25	0.12	0.34	0.88
$p_{\text{ind}}^{\text{bs}}$		0.000	0.000	0.004	0.000	0.000
$p_{\text{time}}^{\text{bs}}$		0.004	0.093	0.109	0.048	0.010

Return Forecast Benchmarking



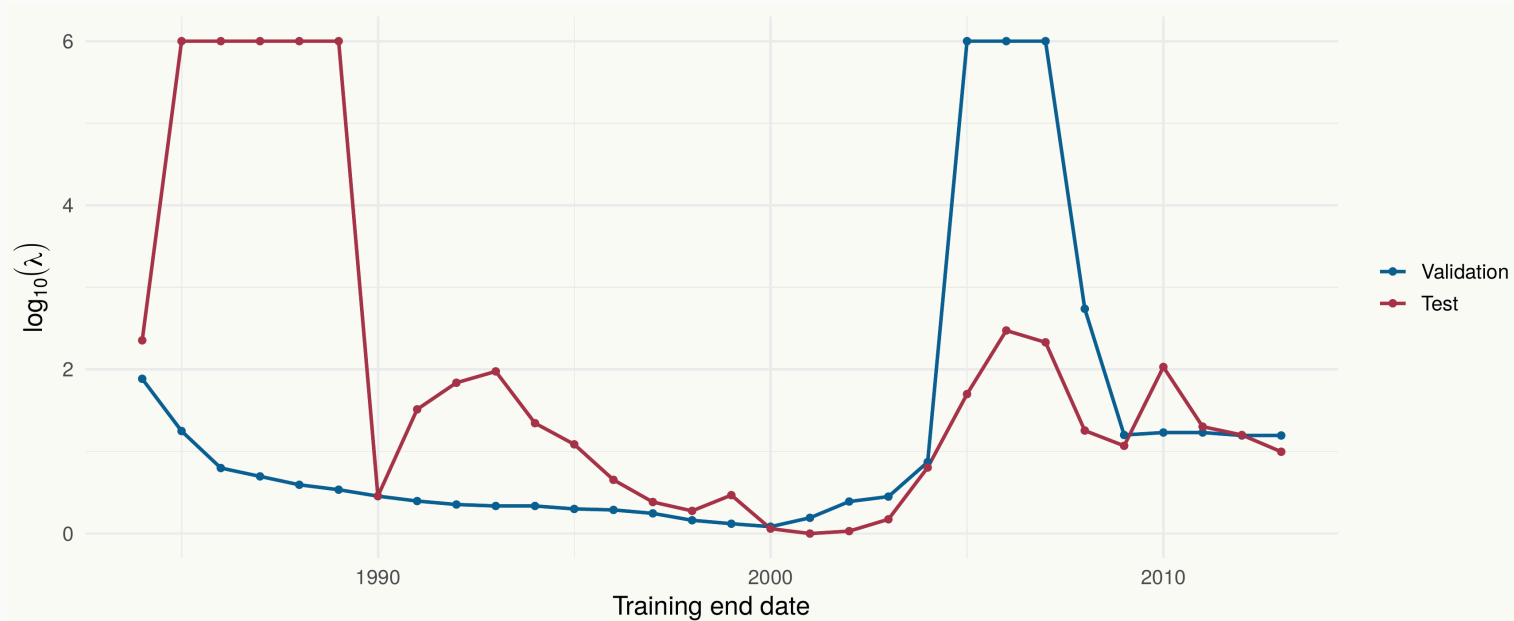
Return Predictability: MCE and MLR

	10-year realized return		
	(i)	(ii)	(iii)
<i>Panel A: Regression estimates</i>			
MCE	0.96*** (0.18)		0.85*** (0.20)
MLR		0.31** (0.13)	0.23** (0.11)
Observations	2,610,475	2,610,475	2,610,475
<i>Panel B: In-sample fit</i>			
MSE	24.14	24.20	24.11
Within R^2 (%)	0.52	0.28	0.67
R^2_{MKT} (%)	3.27	3.05	3.42
<i>Panel C: Out-of-sample fit ($\alpha = 0, \beta = 1$)</i>			
OOS MSE	24.44	25.37	—
OOS R^2_{MKT} (%)	2.10	-1.65	—

Combinations of MCE and MLR

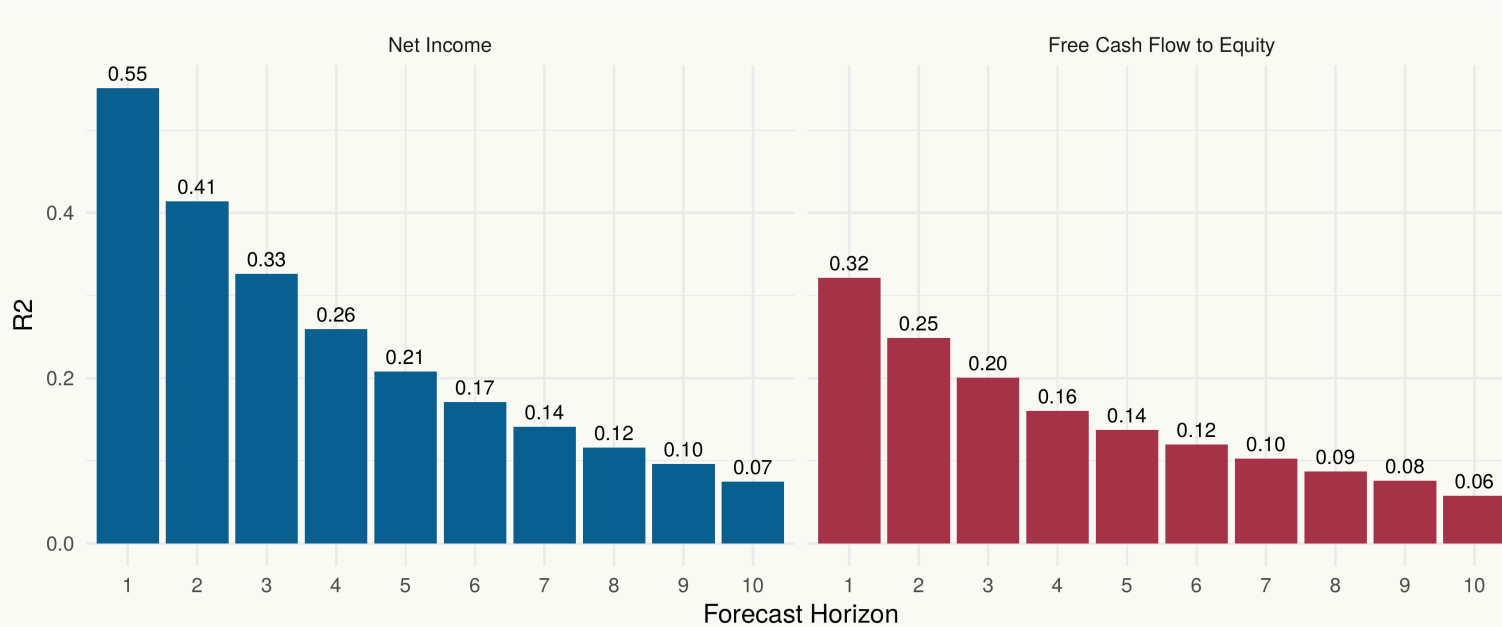
w_{MCE}	$1 - w_{\text{MCE}}$	IS within R^2 (%)	OOS MSE	OOS R^2_{MKT} (%)
0.0	1.0	0.28	25.37	-1.65
0.1	0.9	0.31	25.15	-0.74
0.2	0.8	0.35	24.95	0.05
0.3	0.7	0.40	24.78	0.72
0.4	0.6	0.46	24.64	1.27
0.5	0.5	0.52	24.53	1.71
0.6	0.4	0.59	24.46	2.02
0.7	0.3	0.65	24.41	2.22
0.8	0.2	0.67	24.39	2.30
0.9	0.1	0.62	24.40	2.26
1.0	0.0	0.52	24.44	2.10

Optimal Shrinkage Analysis for Return Prediction



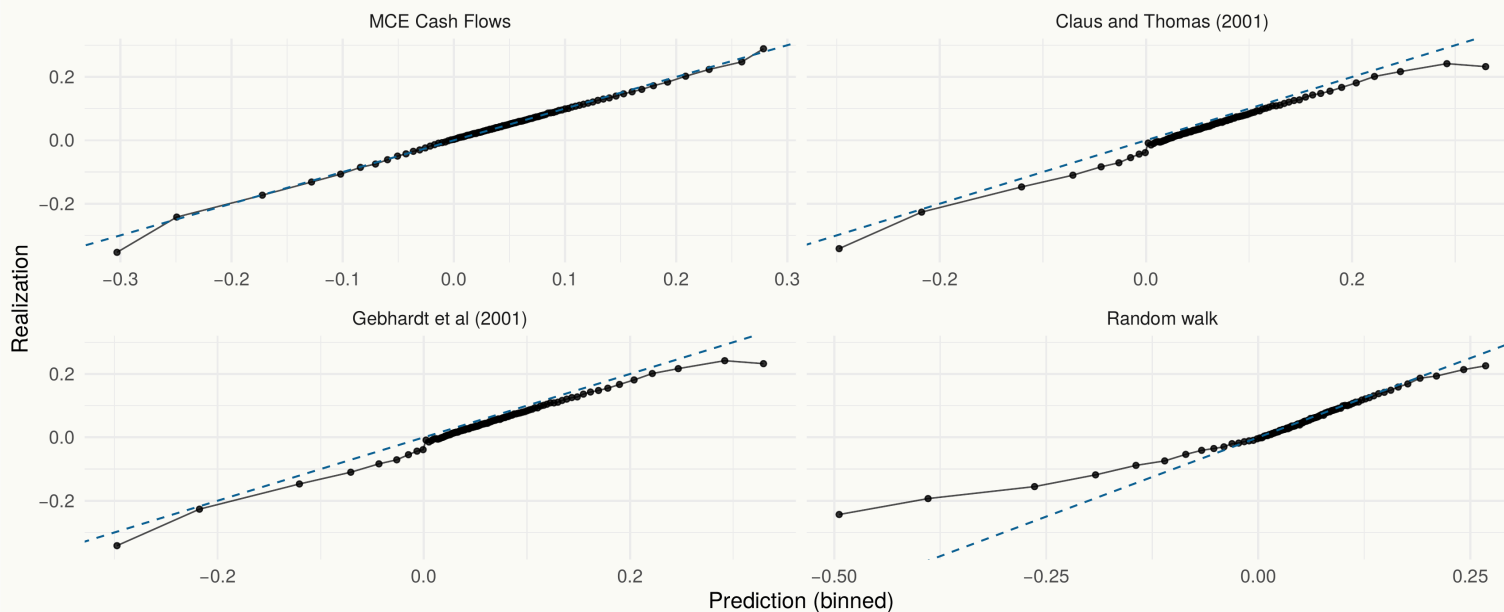
A.3 Cash Flow Predictability

MCE Cash Flow Forecasts (Balanced)



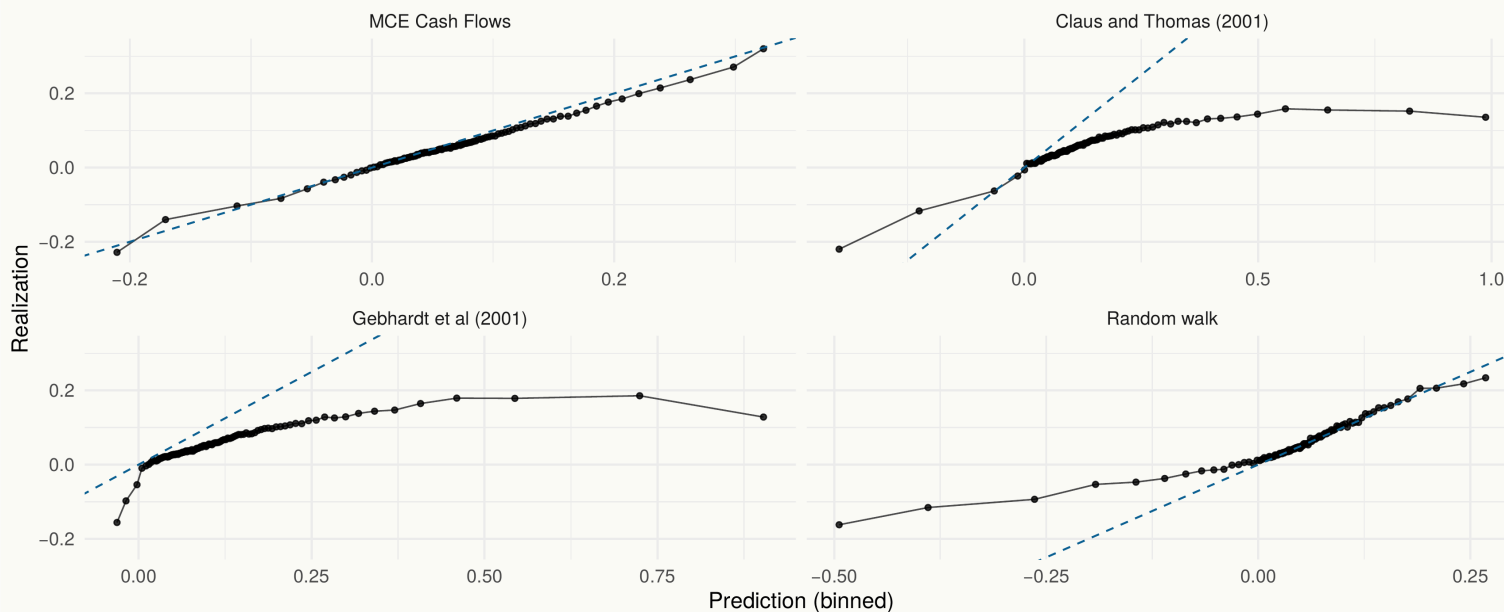
Cash Flow Predictability: Binned Scatters ($h = 1$)

Realized earnings/assets on cash flow predictions, one-year horizon



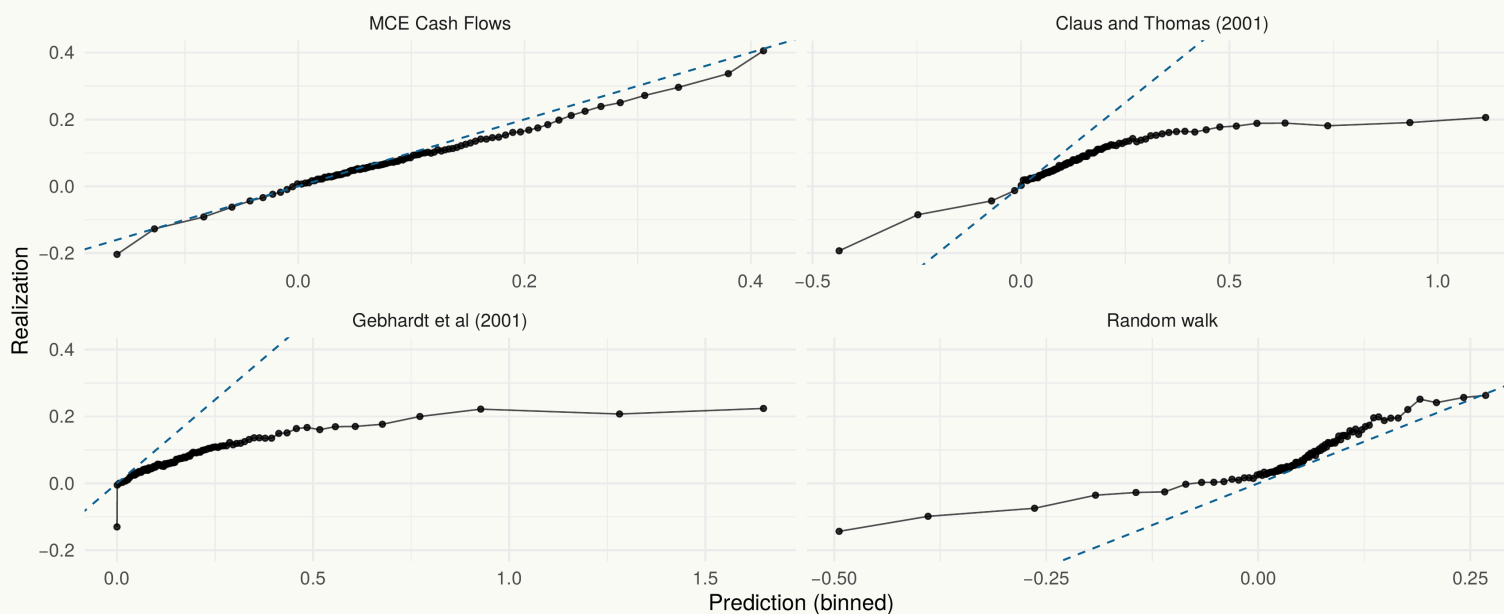
Cash Flow Predictability: Binned Scatters ($h = 5$)

Realized earnings/assets on cash flow predictions, five-year horizon

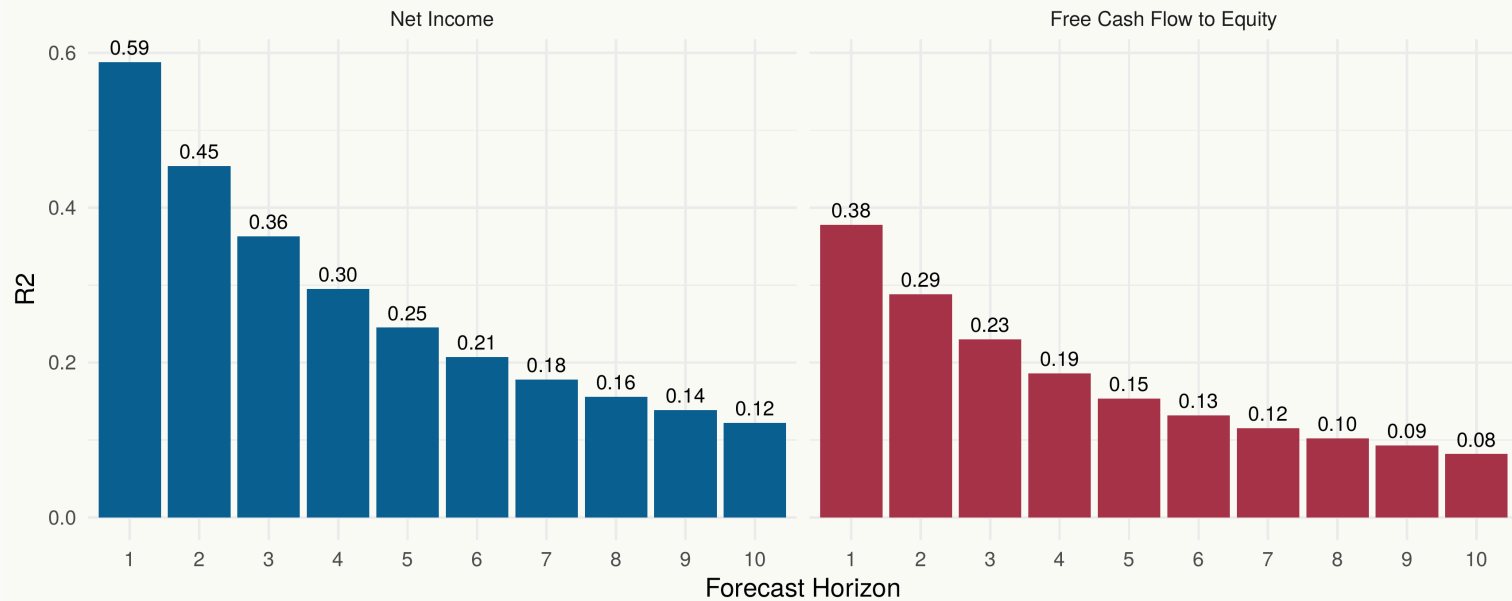


Cash Flow Predictability: Binned Scatters ($h = 10$)

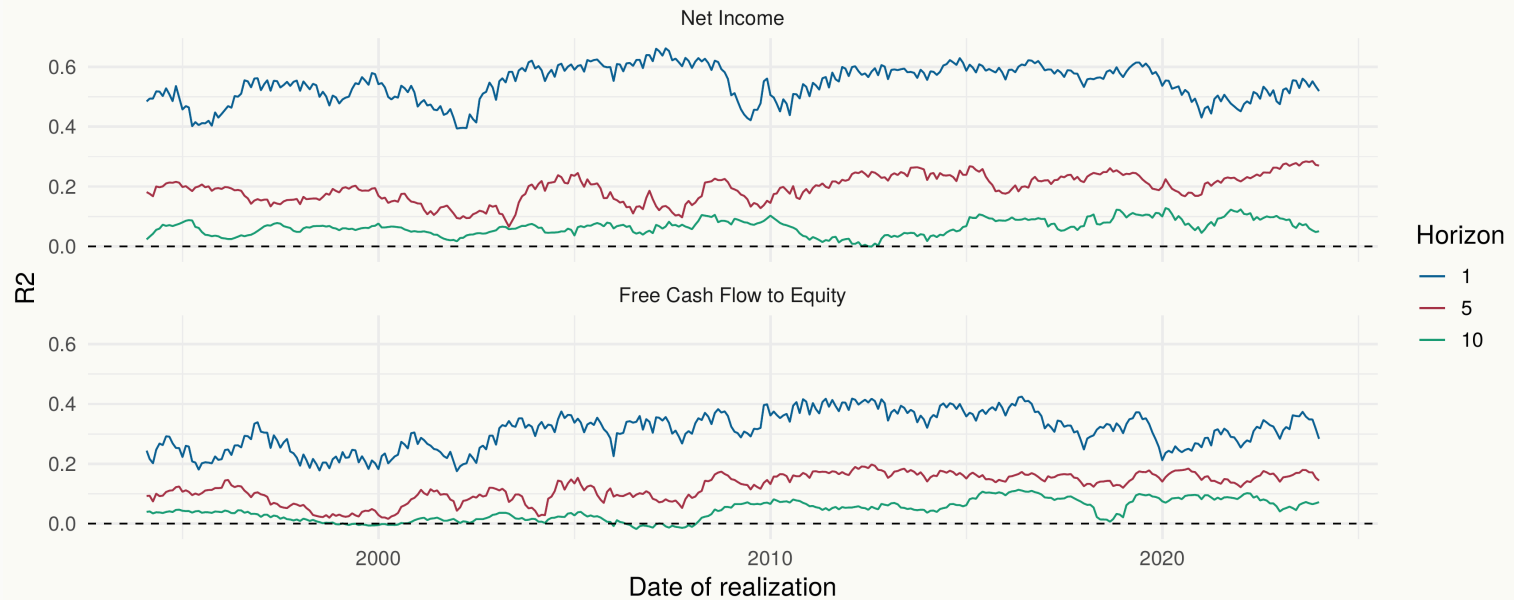
Realized earnings/assets on cash flow predictions, ten-year horizon



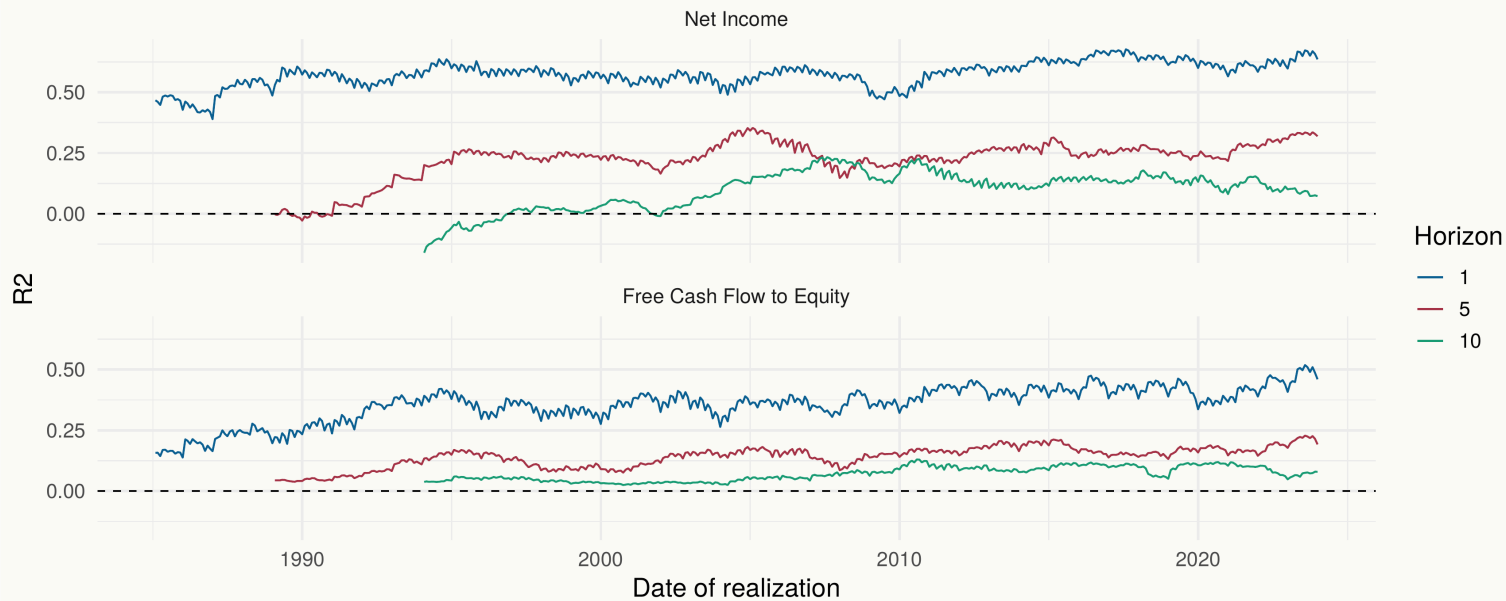
Cash Flow Forecasts (Unbalanced)



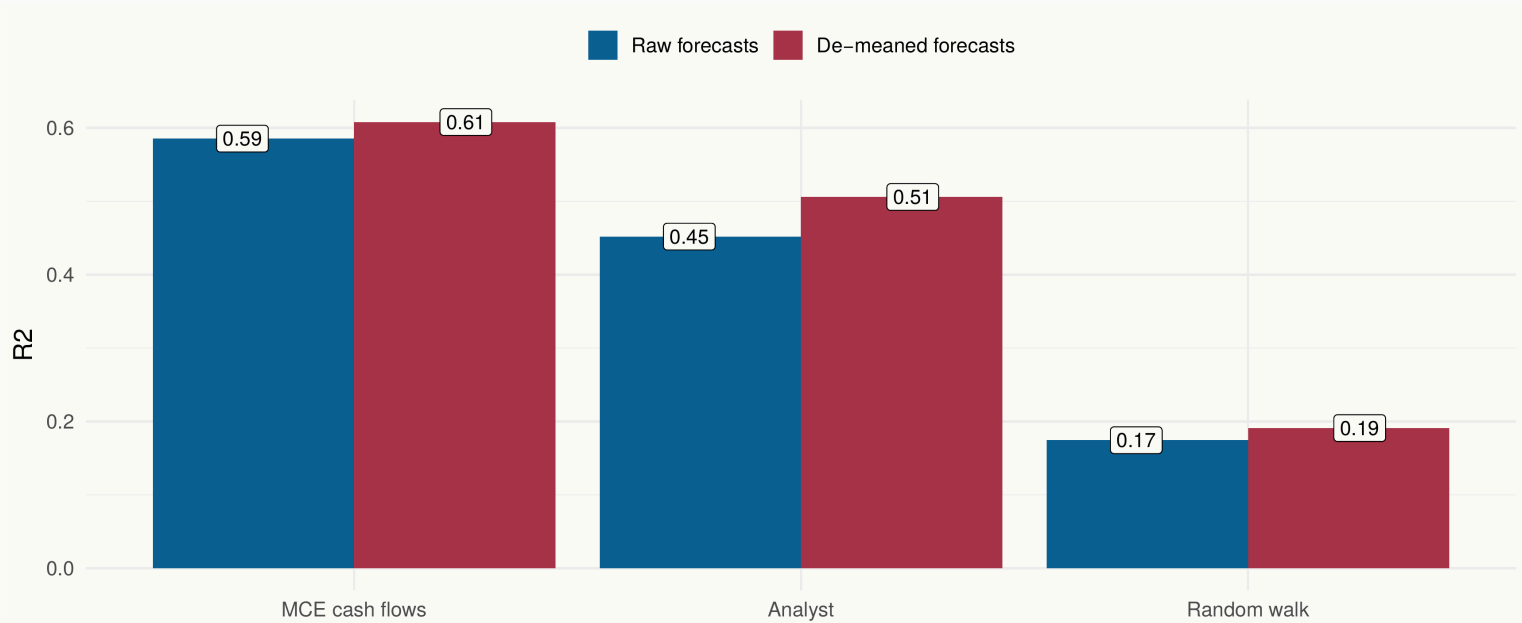
Cash Flow Forecasts over Time (Balanced)



Cash Flow Forecasts over Time (Unbalanced)



ML Forecasts versus Analysts



A.4 Delisting, Imputation, and MCE Validity

Handling Delistings

The problem

- About half of U.S. firms delist within 10 years
- Dropping delisted firms $\Rightarrow$ survivorship bias

Our solution

- Impute returns for delisted firms. Baseline: country-month median
- Robust to other methods:

	U.S.		Global	
	Slope	S.E.	Slope	S.E.
Zero	1.00***	0.11	1.00***	0.17
Risk-free	0.99***	0.10	0.99***	0.16
Median (baseline)	0.93***	0.08	0.96***	0.18
Mean	0.83***	0.06	0.90***	0.17

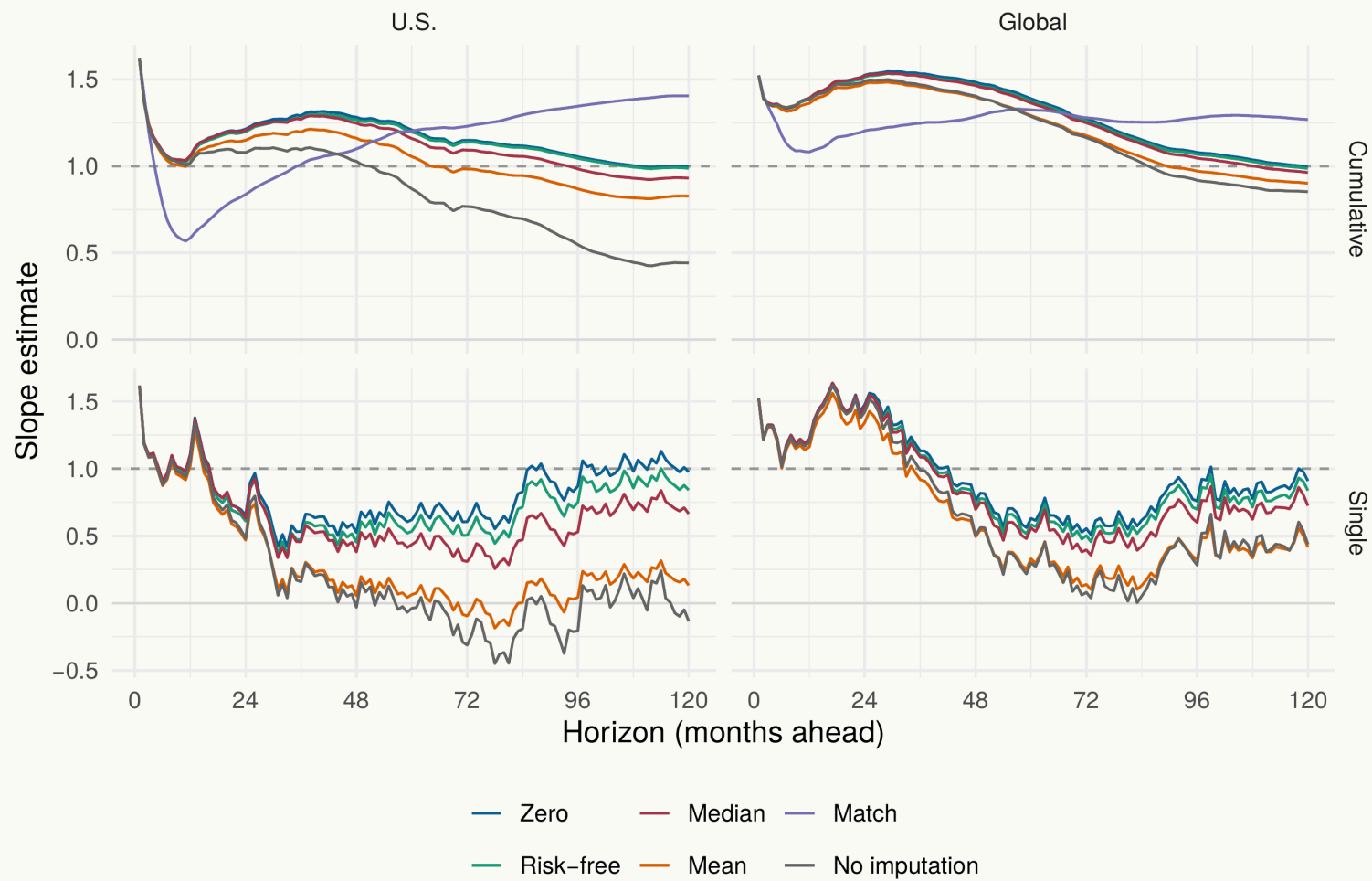
Cumulative Return Predictability: Delisting Robustness

	U.S.			Global		
	Slope	S.E.	<i>N</i>	Slope	S.E.	<i>N</i>
Zero	1.00***	0.11	1,106,464	1.00***	0.17	2,610,489
Risk-free	0.99***	0.10	1,106,464	0.99***	0.16	2,610,489
Median (baseline)	0.93***	0.08	1,106,464	0.96***	0.18	2,610,489
Mean	0.83***	0.06	1,106,464	0.90***	0.17	2,610,489
Match	1.41***	0.10	1,106,464	1.27***	0.12	2,610,489
No imputation	0.44*	0.23	584,079	0.85***	0.24	1,744,048

Single-Period Return Predictability: Delisting Robustness

	U.S.			Global		
	Slope	S.E.	N	Slope	S.E.	N
Zero	0.98***	0.35	1,106,464	0.91***	0.27	2,610,489
Risk-free	0.84**	0.35	1,106,464	0.84***	0.28	2,610,489
Median (baseline)	0.67***	0.23	1,106,464	0.72***	0.25	2,610,489
Mean	0.13	0.26	1,106,464	0.42	0.26	2,610,489
No imputation	-0.13	0.50	584,079	0.44	0.34	1,744,048

Predictability Slope by Horizon Across Delisting Methods



Delisting and Its Reasons by Cost-of-Equity Decile

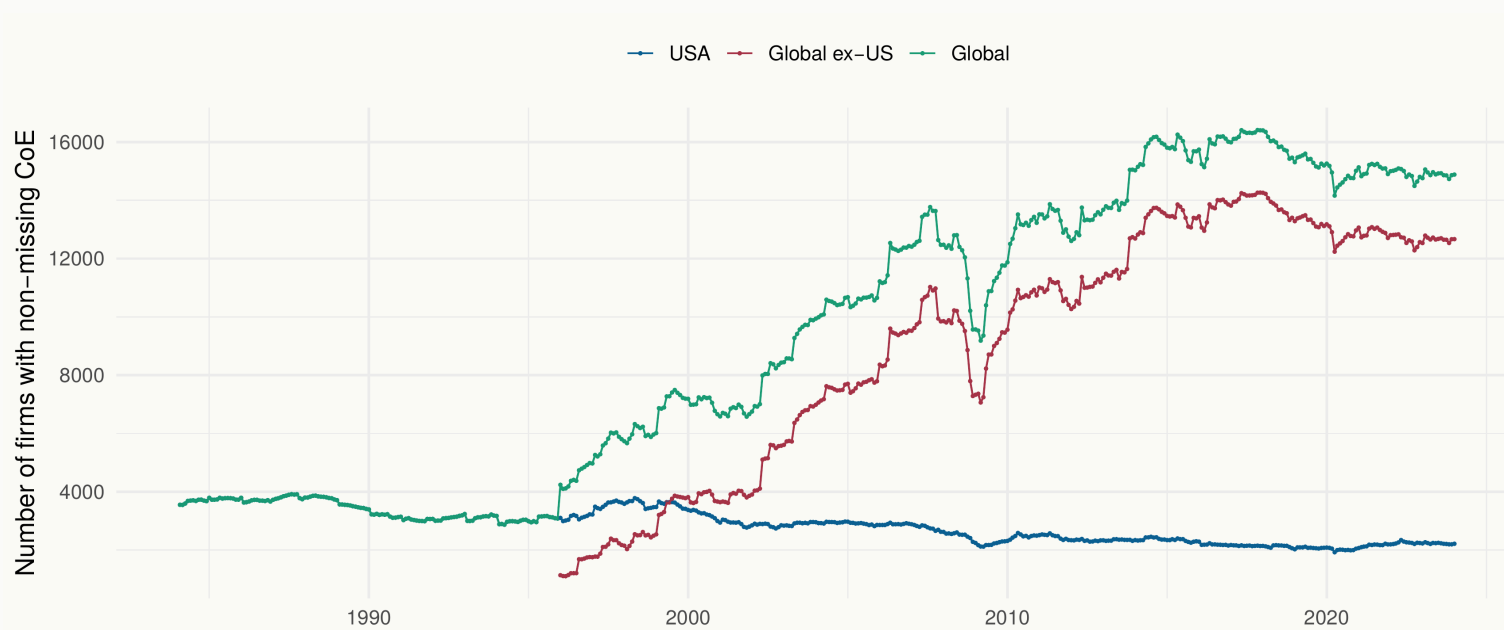
MCE decile	<i>N</i>	Delist 10y %	composition of those delistings (%)			
			Merger	Exchange	Liquidation	Dropped
1 (low)	113,499	62.7	57.8	1.1	0.4	40.7
2	113,298	59.6	65.4	1.4	0.4	32.8
3	113,280	57.9	68.5	1.7	0.3	29.5
4	113,308	55.5	71.0	2.2	0.4	26.5
5	113,343	51.7	72.3	2.7	0.3	24.6
6	113,222	46.5	73.7	2.7	0.4	23.2
7	113,284	41.9	73.9	3.2	0.4	22.5
8	113,314	36.4	74.1	3.2	0.5	22.2
9	113,264	30.7	72.3	3.7	0.5	23.4
10 (high)	113,467	29.1	67.1	3.8	0.7	28.4

Reasons for Missing MCE

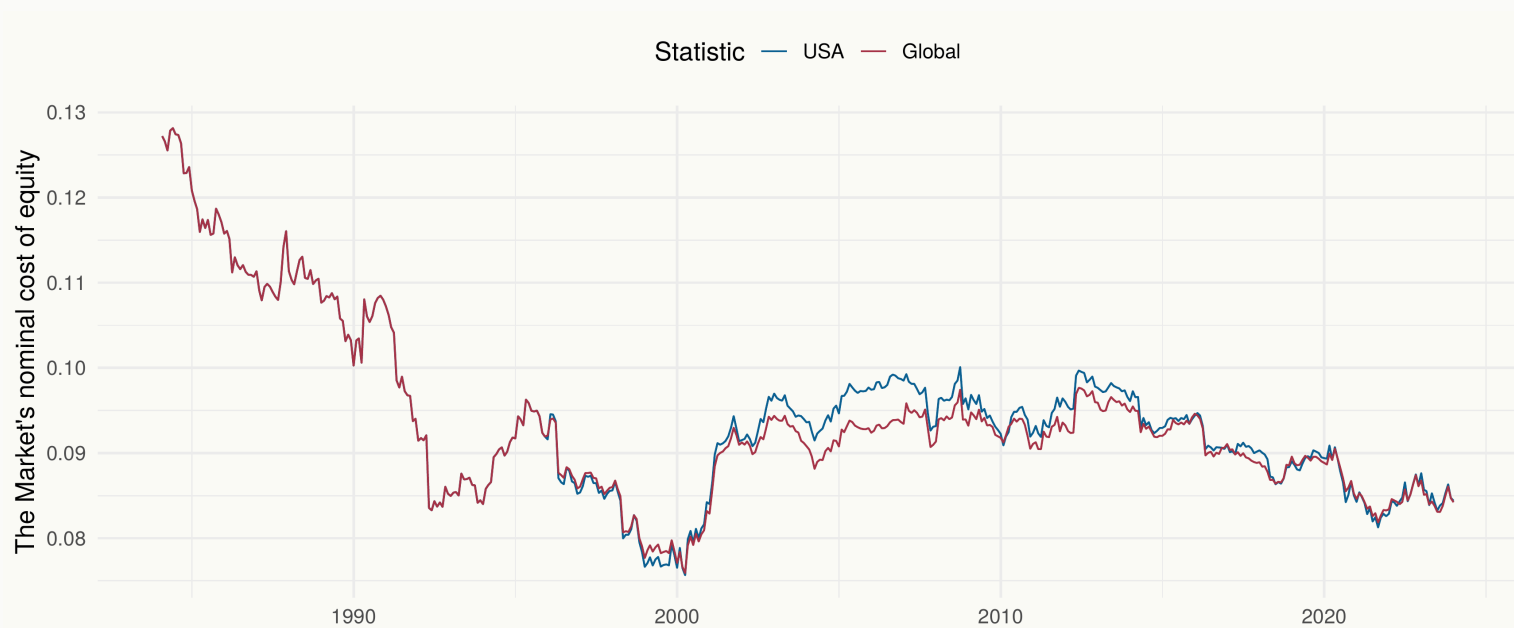
	MCE	MCE (NI)	MCE (CF)
Valid estimate	71.8	59.3	50.2
Implied rate below terminal growth rate	18.0	35.1	24.2
Implied rate above 10% real-rate cap	4.2	5.6	25.6
Both input estimates invalid	6.0	—	—
Other	<0.1	<0.1	<0.1
Observations (millions)	6.41	6.41	6.41

A.5 Cost of Capital Estimates

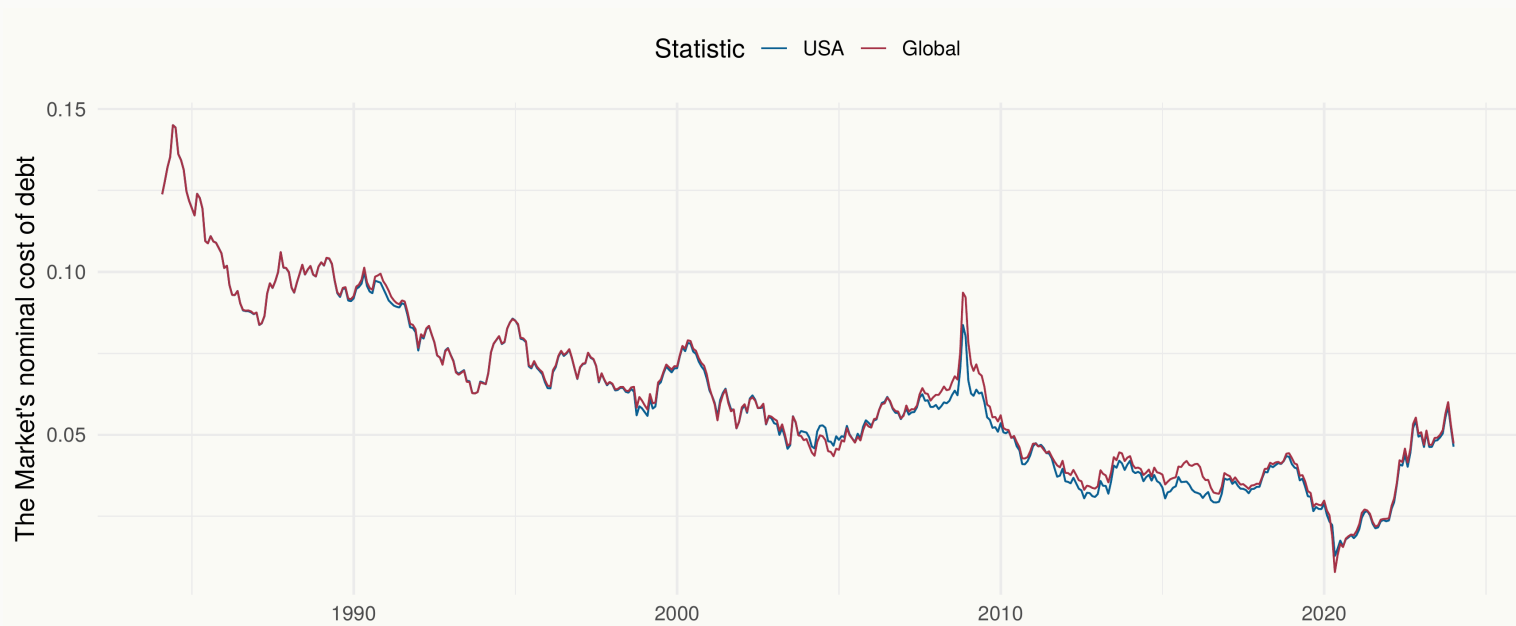
MCE Coverage



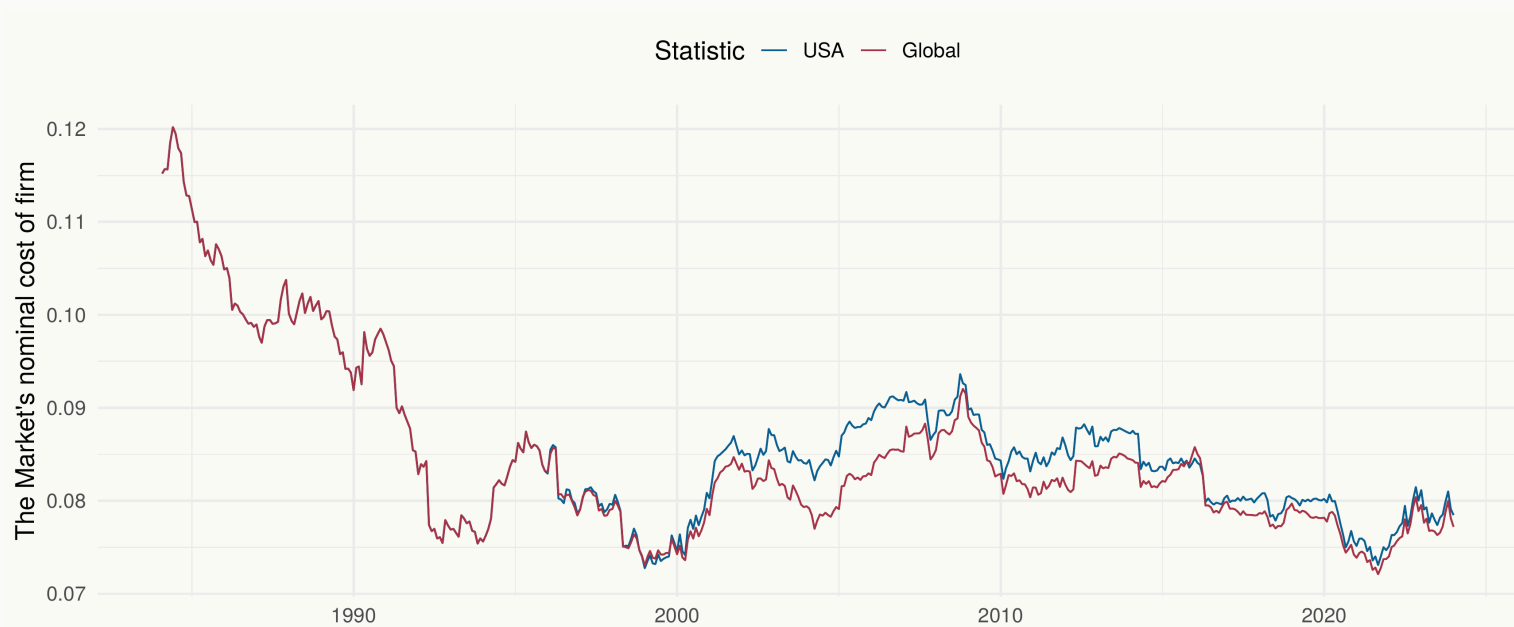
Time Variation in the Market's Nominal MCE



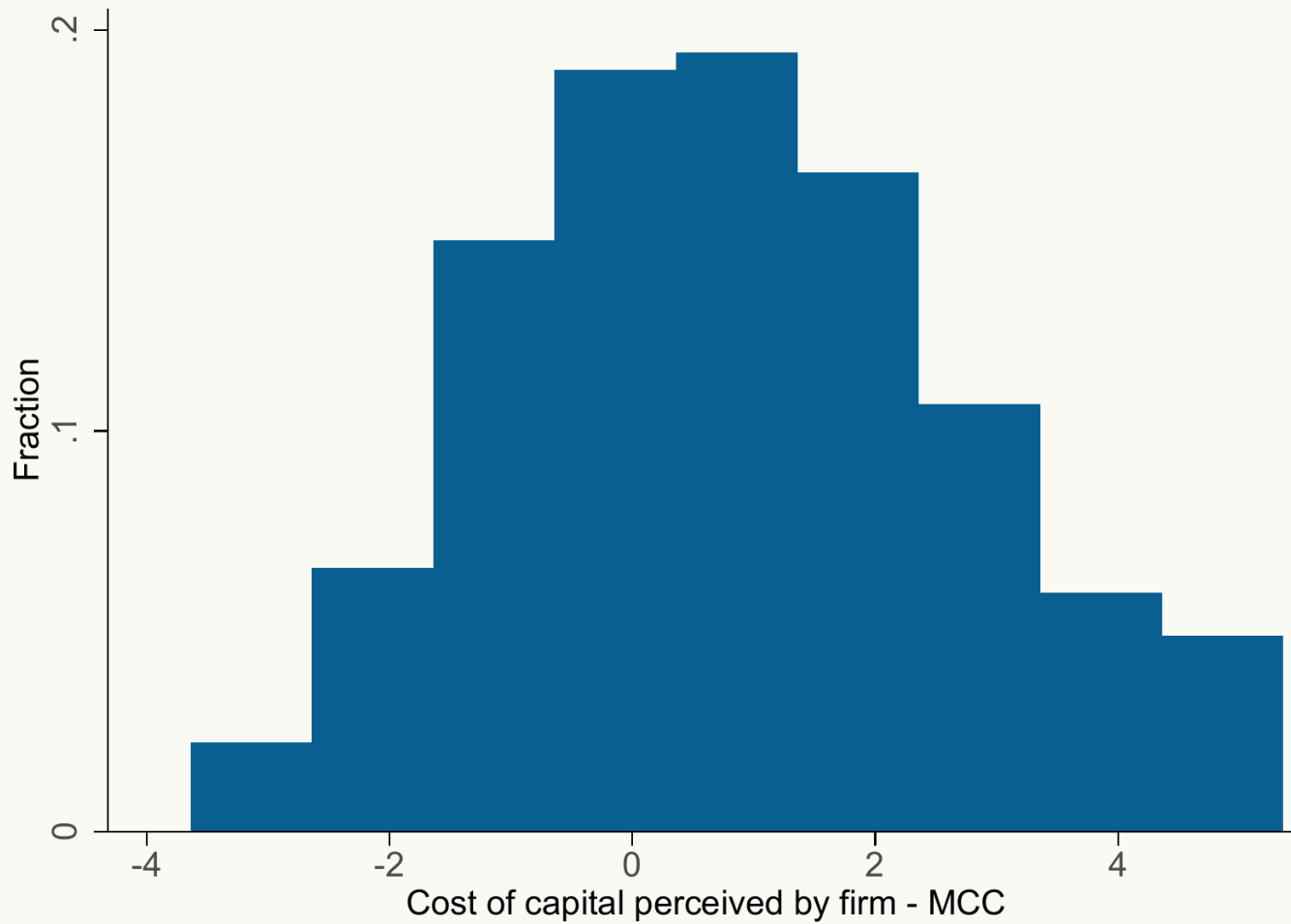
Time Variation in the Market's Nominal MCD



Time Variation in the Market's Nominal MCC



Histogram of Deviations from the MCC



A.6 Risk or Mispricing

Risk or Mispricing: Cost of Debt

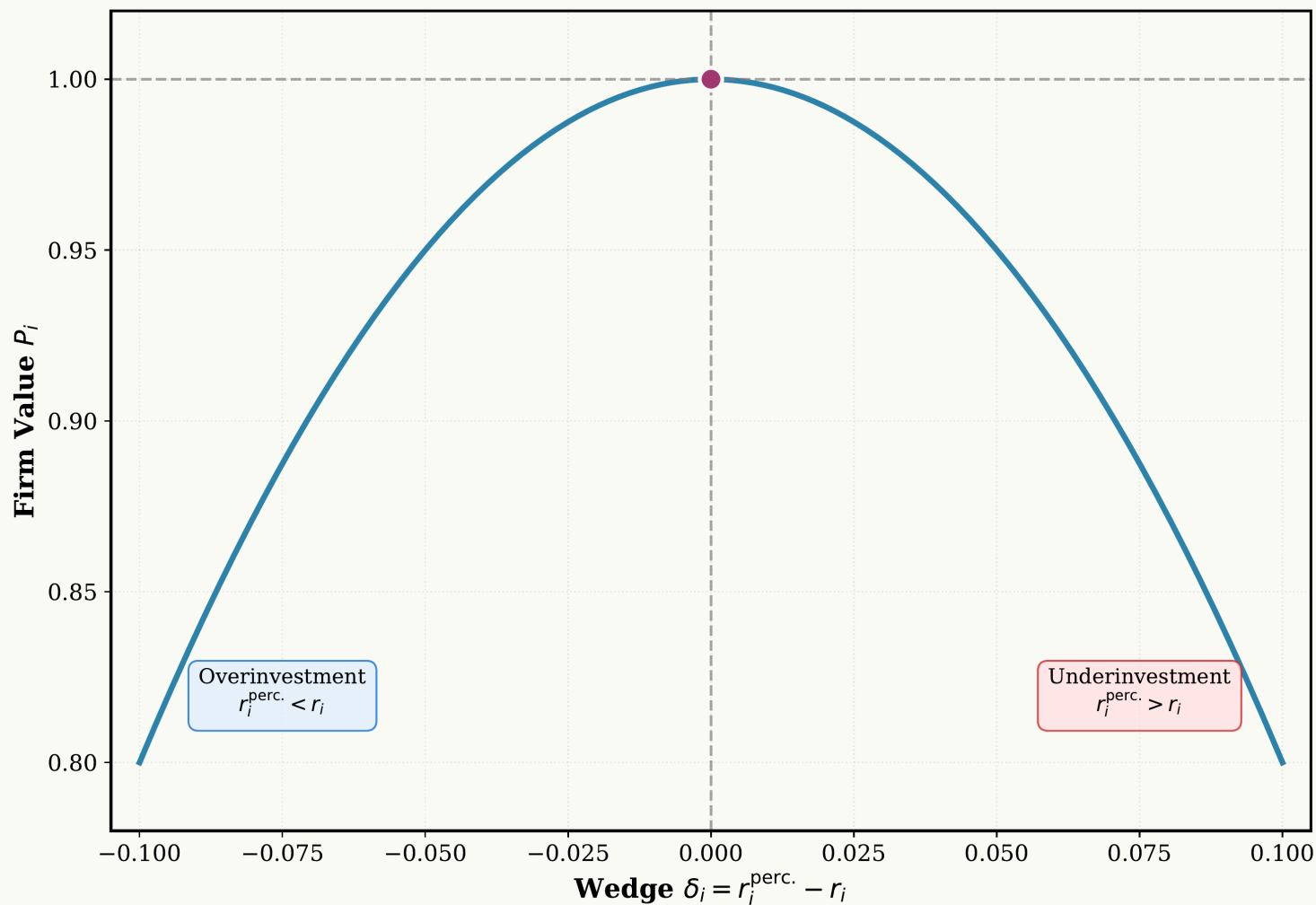
	Risk			Mispricing		
	Morningstar	Value Line	Beta	EAR	FE	$\mathbb{E}[\text{FE}]$
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
MCD	0.08*** (0.02)	0.09*** (0.01)	0.05*** (0.01)	-0.01 (0.01)	-0.04 (0.03)	-0.21*** (0.04)
Observations	113,471	383,210	376,073	212,169	27,027	27,027
R ²	0.016	0.018	0.008	0.003	0.026	0.060

Risk or Mispricing: Cost of Capital

	Risk			Mispricing		
	Morningstar	Value Line	Beta	EAR	FE	$\mathbb{E}[\text{FE}]$
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
MCC	-0.13*** (0.02)	-0.24*** (0.01)	-0.09*** (0.01)	0.07*** (0.01)	0.32*** (0.05)	0.56*** (0.08)
Observations	102,848	349,750	343,788	162,795	21,273	21,273
R ²	0.040	0.125	0.019	0.004	0.045	0.099

A.7 Firm Value and Investment

Model: Inverted U Between Market Value and MCC Deviations



Inverted U: Market Value and Deviations from the MCC

	(1)	(2)	(3)	(4)	(5)
	U.S.	Global	≤ -0.3	≥ 0.3	$> -0.3, < 0.3$
Perc. CoC — MCC	-0.081*** (0.019)	-0.096*** (0.018)			
Perc. CoC — MCC			0.087* (0.049)	-0.070*** (0.025)	-0.173 (0.210)
Observations	3,098	5,204	1,509	3,055	640
Within R ²	0.0185	0.0266	0.00924	0.0154	0.00240

Robustness: Stock Returns and Investment Announcements

	(1)	(2)	(3)
	Global	Global	U.S.
(Perc. CoC — MCC) > 0	0.199** (0.098)	0.142* (0.084)	0.170** (0.084)
Observations	709	709	709
Within R ²	0.0107	0.00680	0.00983

A.8 Model Details

Predicting Cash Flows (II)

- The model, $f(\cdot)$: LightGBM — a gradient-boosted decision tree ensemble ([Ke et al. 2017](#))
 - State of the art for (big) tabular data ([Shwartz-Ziv and Armon 2022](#))
 - Natively handles non-linearities, interaction effects, and missing inputs
- The input data, X_t^i
 - 382 firm characteristics based on accounting, price, analyst, industry, and country data
 - Based on [Jensen, Kelly, and Pedersen \(2023\)](#)
- The target, y_{t+h}^i
 - Net income (NI) and free cash flow to equity
$$\text{FCFE} = \text{NI} + \text{DP} - \text{CAPX} - \Delta\text{NWC} + \Delta\text{Debt}$$
 - Adjusted for inflation
 - Estimate separate models for $t + 1, \dots, t + 10$

Hyperparameter Search Space

Parameter	Range	Scale
feature_fraction	$[1/P, 1]$	Linear
num_leaves	$[1, 127]$	Linear
min_data_in_leaf	$[1, 500]$	Linear
bagging_fraction	$[0.2, 1.0]$	Linear
lambda_l2	$[10^{-4}, 10^5]$	Log

LightGBM models tuned over a 20-point max-entropy space-filling design; P is the number of features. `lambda_l1 = 0` and `max_depth = -1` held fixed.